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1 Introduction

Crop classification and monitoring are both central research topics in modern agriculture. Although it may appear straightforward, accurately identifying which crops are growing in specific regions is essential for proper agricultural management and food security. Different crops require distinct environmental conditions to achieve optimal and healthy growth, including temperature, soil composition, elevation, and moisture.

The emergence of satellites in the 20th century improved this field tremendously. Satellite imagery enables continuous observation of crop development across large landscapes throughout extended periods of time, favouring both visualisation and in-depth analysis of crop growth stages and distribution patterns.

More recently, the rise of deep learning has further enhanced crop classification methodologies. By leveraging significant advances in computational power and data availability, it has enabled the development of powerful tools and advanced architectures that have substantially improved classification accuracy.

Among the most impactful deep learning architectures of the past decade are Convolutional Neural Networks (CNNs), which have revolutionised image analysis tasks, and Transformers, which use attention mechanisms to efficiently capture complex relationships within large-scale datasets.

However, despite these major improvements, deep learning cannot always perfectly predict crop types. Achieving high classification accuracies requires carefully balancing dataset size and quality against model architecture complexity, in order to mitigate the risk of models being either too simple or too complex to process the given data – a failure that manifests as underfitting or overfitting. Finding an adequate middle-ground is therefore essential.

With the goal of exploring the capacity of deep learning models for crop classification, this report presents three main components:

- **Part 1:** Replication of an existing CNN–Transformer-based architecture (MCTNet), which takes as input encoded satellite images of crops in the states of Arkansas and California (United States) and predicts crop types after a training phase, in order to evaluate the effectiveness of deep learning for crop classification.

- **Part 2:** Enrichment of the initial satellite image dataset with additional environmental features – including climate, soil, moisture, and topography – followed by a re-evaluation of the replicated MCTNet model's performance with this supplementary information, to assess the impact of these variables on crop prediction accuracy.

- **Part 3:** Design of an improved model architecture aimed at maximising performance and crop classification accuracy.

These three components build upon a shared baseline drawn from the work of Wang et al. [1], published in *Computers and Electronics in Agriculture* (2024). Their study introduces MCTNet, a lightweight multi-stage CNN–Transformer network for pixel-based crop mapping from time-series Sentinel-2 imagery, evaluated on two U.S. study areas: Arkansas (large-scale annual crops) and California (a more fragmented, crop-diverse landscape). Their datasets, preprocessing protocol, and class definitions serve as the direct foundation for our work: we reproduce their model, enrich their data with environmental covariates, and propose architectural improvements upon their baseline.

PART 1: Base Classification MCTNet Model

2 Description and Exploration of Datasets & Preprocessing Pipeline

2.1 Description and Exploration of Datasets

Dataset size and origin. In order to remain faithful to the original MCTNet study, two datasets are used: one for the state of Arkansas and one for California. Both are derived from Sentinel-2 satellite imagery for the year **2021**. Each dataset contains **10,000 samples**, where each sample represents a single agricultural pixel observed over 36 time steps at 10-day intervals, providing full annual coverage of the growing season.

Spectral features. Each sample is represented as a tensor $X \in R^{36\times10}$, where the 36 rows correspond to temporal acquisitions and the 10 columns correspond to the Sentinel-2 spectral bands: Blue, Green, Red, Red-Edge 1–4, NIR, SWIR1, and SWIR2. These bands span the electromagnetic spectrum from the visible range (Blue: $\sim$490 nm) to the short-wave infrared (SWIR2: $\sim$2190 nm), providing sensitivity to vegetation biochemistry, water content, and canopy structure. Alongside the spectral features, a binary cloud mask $m \in \{0, 1\}^{36}$ indicates which time steps contain valid, cloud-free observations.

The main properties of both datasets are summarised in Table 1.

Table 1: Main characteristics of the datasets used in this study.

Dataset Characteristic	Arkansas Dataset	California Dataset
Number of samples	10,000	10,000
Satellite source	Sentinel-2	Sentinel-2
Spectral bands used	10 (B2–B8A, B11, B12)	10 (B2–B8A, B11, B12)
Temporal resolution	10-day intervals	10-day intervals
Number of time steps	36	36
Features per sample	360	360
Cloud mask information	`valid_t*` included	`valid_t*` included
Data format	CSV time-series	CSV time-series
Input representation	Multi-temporal sequences	Multi-temporal sequences
Geographical coverage	Arkansas, USA	California, USA

Crop classes. Arkansas contains 5 crop classes: Soybeans (4,704 samples), Rice (2,488), Corn (1,392), Cotton (806), and Others (610). California contains 6 crop classes: Grapes (2,008), Rice (1,856), Alfalfa (985), Almonds (808), Pistachios (585), and Others (3,758).

A detail worth noting is that both datasets exhibit significant class imbalance. In the California dataset, the Others category represents 37.6% of all samples, whereas Pistachios account for only 5.85%. Such imbalance makes the classification of minority classes particularly challenging, and motivates the use of balanced class weighting during training (see Section 3.3).

The crop distribution in each state is illustrated in Figure 1.

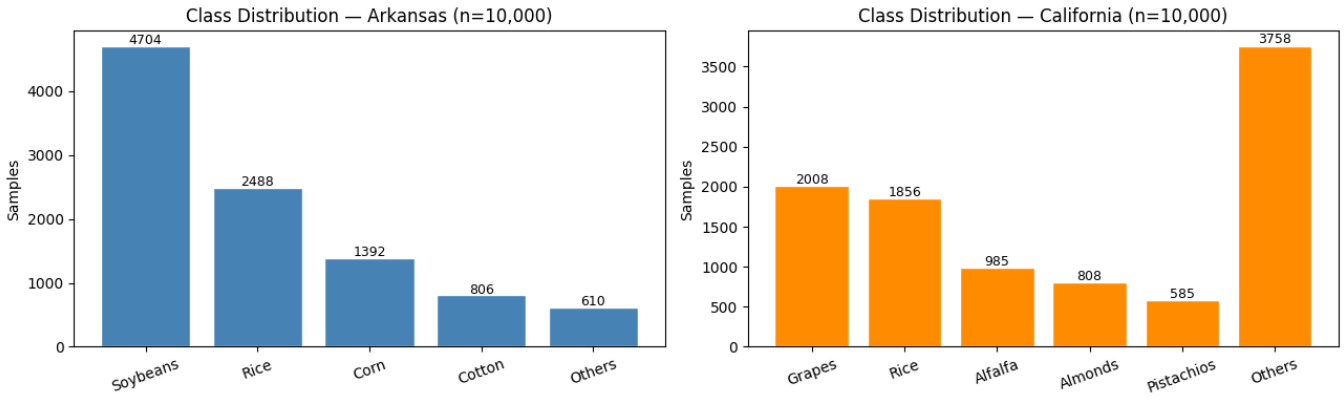


Figure 1: Crop distribution in Arkansas and California.

NDVI temporal profiles. The Normalized Difference Vegetation Index (NDVI = $(NIR-Red)/(NIR+Red)$) provides a crop-discriminating feature that can be computed directly from the Sentinel-2 bands. NDVI time series reveal distinctive phenological signatures for each crop type: in Arkansas, rice exhibits a sharp NDVI increase in late spring corresponding to paddy flooding and transplanting, while soybeans show a broader summer peak. In California, tree crops (almonds, pistachios, grapes) maintain relatively high year-round NDVI with a smoother seasonal arc compared to annual crops such as rice and alfalfa. Cloud-masked time steps are treated as missing (NaN) for visualisation purposes, but are preserved as zero in the input tensor.

Figure 2 showcases the NDVI values of each crop type throughout the year in both states.

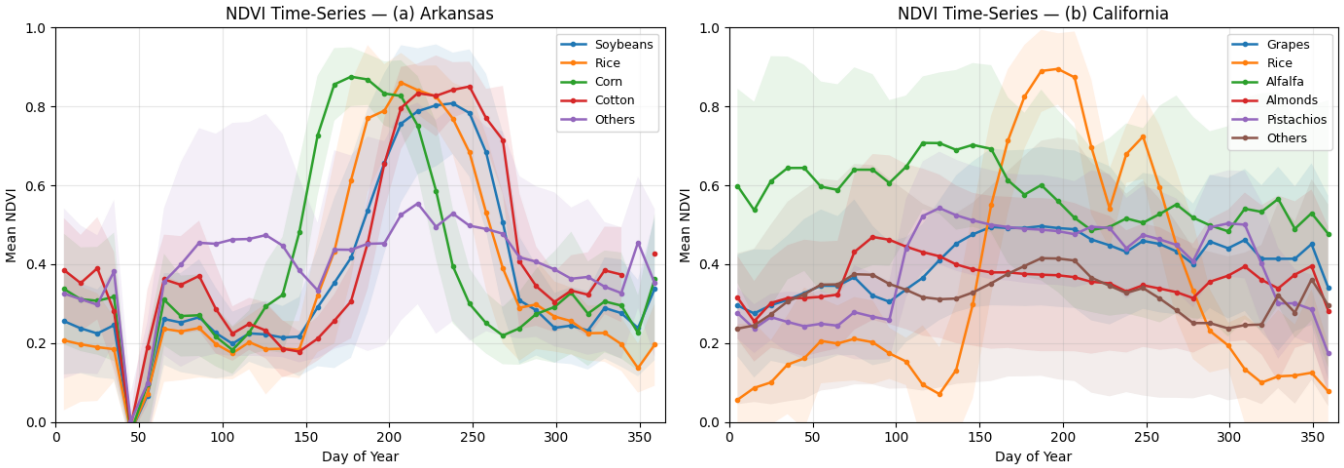


Figure 2: NDVI distribution in Arkansas and California.

2.2 Preprocessing Pipeline

Preprocessing represents an essential step that must be performed before training in order for the model to correctly interpret the data. The preprocessing pipeline is applied consistently across all three parts of this project, with additional extensions introduced in Parts 2 and 3 to accommodate the new features and improvements brought by those sections. Part-specific preprocessing modifications – covariate normalisation in Part 2, and cloud-gap interpolation and vegetation index computation in Part 3 – are described in their respective sections.

Missing data handling. Cloud-masked observations (where $\texttt{valid_t}_k = 0$) are **left as zero** in the spectral tensor. This zero-padding is not simply a filler: the binary cloud mask $m \in \{0,1\}^{36}$ is passed as a separate input to the ALPE module, which uses it to modulate the sinusoidal positional encoding so that the model learns to suppress the contribution of zero-padded positions automatically. Consequently, masked time steps do not need to be removed or imputed at the preprocessing stage – the architecture itself handles their reduced reliability.

Train/validation/test split. A stratified split is applied per crop class: **300 samples per class** are reserved for training and validation (240 train / 60 validation), and the remainder forms the test set ($\approx$ 8,200 samples per state). The split is performed *before* any normalisation step to prevent data leakage: computing statistics over the full dataset and then splitting would allow the model to indirectly access validation and test information during training, artificially inflating reported performance.

Spectral normalisation. Each of the 10 spectral bands is independently rescaled to the $[0,1]$ range using min–max normalisation:

$$\hat{x}_{t,b} = \frac{x_{t,b} - v_{\min,b}}{v_{\max,b} - v_{\min,b}} \tag{1}$$

where $v_{\min,b}$ and $v_{\max,b}$ are the minimum and maximum values of band b computed *on the training split only*. The same scaler is subsequently applied, without refitting, to the validation and test splits. This is a critical implementation detail: an earlier version of the pipeline erroneously computed the scaler on the full dataset before splitting, which constituted a data leakage bug (FIX-1 in Appendix B). Cloud-masked positions, stored as zero in the input tensor, remain as zero after normalisation since their band values would fall within the training range.

Class weighting. To counteract the effect of class imbalance, balanced class weights are applied during training. For each class c, the weight is defined as:

$$w_c = \frac{N}{K \cdot n_c} \tag{2}$$

where N is the total number of training samples, K is the number of classes, and n_c is the number of training samples in class c. These weights are passed to the cross-entropy loss so that the model receives proportionally stronger gradient updates from underrepresented classes such as Cotton (Arkansas) and Pistachios (California).

Summary. Table 2 provides a concise reference for all base preprocessing steps and their order of application. Steps marked as Part-specific are described in the sections where they are introduced.

Table 2: Summary of the base preprocessing pipeline (Parts 1–3). Part-specific extensions are described separately in Sections 5.4 and 4.2.

Step	**Operation**	**Details**
1	Stratified train/val/test split	240/60 train/val per class; remainder as test. Performed before any normalisation.
2	Spectral min–max normalisation	Per-band, fitted on train split only. Applied identically to val and test.
3	Cloud mask zero-padding	Masked positions left as zero; mask m passed separately to ALPE.
4	Class weight computation	Balanced weights $w_c = N/(K \cdot n_c)$ applied to cross-entropy loss.
–	Covariate z-score standardisation	Part 2 only. StandardScaler fitted on train split.
–	Cloud-gap linear interpolation	Part 3 only. Gaps of ≤ 2 consecutive steps interpolated.
–	Vegetation index computation	Part 3 only. NDVI, IRECI, MTCI, S2REP appended per time step.

3 Reproduction of the Baseline Proposed MCTNet Model

3.1 Overview of the MCTNet Study

MCTNet (Multi-source Crop Type recognition Network) was proposed by Wang et al. [1] as a lightweight architecture for pixel-level crop mapping from Sentinel-2 time series. Its core innovation is the combination of three parallel CNN–Transformer stages within a hierarchical feature extraction pipeline, coupled with an Attention-based Learnable Positional Encoding (ALPE) that adapts sinusoidal positional encodings to per-pixel cloud-gap patterns. The model is deliberately compact to enable deployment over large geographic areas.

3.2 Architecture

The MCTNet architecture processes two inputs simultaneously: the spectral time series $X \in R^{36\times10}$ and the cloud mask $m \in \{0,1\}^{36\times1}$. The full dimensional trace is:

Stage	**Output shape**
Input (spectral + mask)	$(N, 36, 10)$ and $(N, 36, 1)$
Stage 1: CNN(10) ‖ Transformer+ALPE(10) → Concat	$(N, 36, 20)$
MaxPool(2)	$(N, 18, 20)$
Stage 2: CNN(20) ‖ Transformer(20) → Concat	$(N, 18, 40)$
MaxPool(2)	$(N, 9, 40)$
Stage 3: CNN(40) ‖ Transformer(40) → Concat	$(N, 9, 80)$
GlobalMaxPool	$(N, 80)$
Dense(softmax)	(N, K)

Each stage contains a **CTFusion** module, which runs a CNN branch and a Transformer branch in parallel on the same input, then concatenates their outputs. This concatenation – rather than addition – is key: it doubles the channel count at each stage, allowing the two branches to learn complementary representations without interference.

CNN sub-module. Each CNN branch implements a two-layer 1D residual block with Batch Normalisation:

$$h = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{k=3}(x)) + \mathrm{BN}(\mathrm{Conv}_{k=3}(\mathrm{Conv}_{k=3}(x)))) \tag{3}$$

A 1×1 projection is applied to the skip connection whenever the channel dimension changes between stages.

Transformer sub-module. The Transformer follows the pre-LN formulation with Multi-Head Self-Attention (MHSA) using $n_{\text{heads}} = 5$ and key dimension $d_k = d/n_{\text{heads}}$:

$$z = \mathrm{LayerNorm}(x + \mathrm{MHSA}(x)) \tag{4}$$

$$h = \mathrm{LayerNorm}(z + \mathrm{FFN}(z)) \tag{5}$$

A 1×1 convolution projects the Transformer output to match the CNN branch's filter count before concatenation.

ALPE module. Standard sinusoidal positional encoding is agnostic to data quality. ALPE addresses this by masking the base encoding at cloud-contaminated positions and applying channel-wise attention (ECA) to produce a cloud-aware position representation (Wang et al., Eq. 3):

$$\mathrm{PE}_{\mathrm{masked}} = m \odot \mathrm{PE}_{\mathrm{sin}} \tag{6}$$

$$a = \sigma\left(\mathrm{Conv}_{k=3}\left(\frac{1}{T}\sum_{t}\mathrm{Conv}_{k=3}(\mathrm{PE}_{\mathrm{masked}})\right)\right) \tag{7}$$

$$\mathrm{ALPE}(m) = \mathrm{Conv}_{k=3}(\mathrm{PE}_{\mathrm{masked}}) \odot a \tag{8}$$

The ALPE output is added to the spectral features *only at Stage 1*, injecting temporally-aware context that signals where observations are missing.

Figure 3 shows the complete MCTNet architecture diagram.

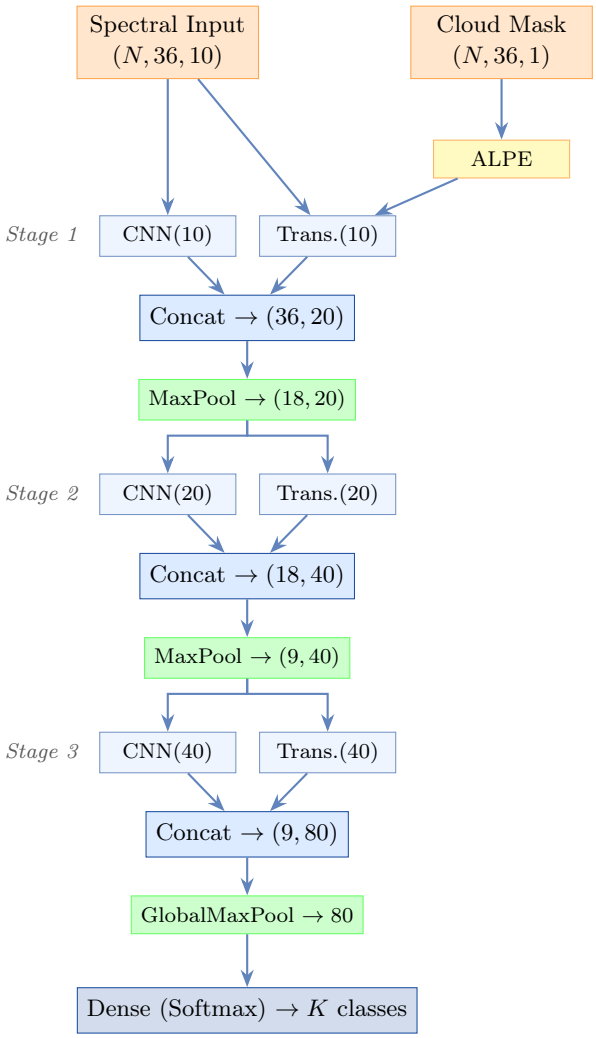


Figure 3: Baseline MCTNet architecture.

3.3 Training Settings

The model is implemented in TensorFlow 2.x / Keras following the hyperparameters reported in the original MCTNet study. Table 3 summarises the training configuration used in Part 1.

Table 3: Training hyperparameters for the baseline MCTNet reproduction (Part 1).

Parameter	**Value**
Optimizer	Adam
Learning rate	10^{-3}
Batch size	32
Maximum epochs	200
Early stopping	patience 20, `restore_best_weights=True`
LR scheduler	ReduceLROnPlateau (factor 0.5, patience 8, min 10^{-6})
Train / val per class	240 / 60
Transformer heads	5
CNN kernel size	3
Class weighting	Balanced ($w_c = N/(K \cdot n_c)$)

3.4 Results and Discussion

Table 4 reports the test-set results for both states, comparing our Part 1 reproduction to the original MCTNet study's published metrics. Overall Accuracy (OA), Cohen's Kappa, and macro-averaged F1-score are reported.

Table 4: Part 1 baseline reproduction results compared with the original MCTNet study. Training convergence curves and confusion matrices for both states are provided in Appendix D.1.

	Arkansas			**California**		
Model	OA	κ	F1	OA	κ	F1
Wang et al. (MCTNet)	0.968	0.951	0.933	0.852	0.806	0.829
Our reproduction	0.937	0.905	0.885	0.791	0.730	0.743
Gap	−0.031	−0.046	−0.048	−0.061	−0.076	−0.086

Our reproduction achieves an OA of 0.937 in Arkansas and 0.791 in California, falling short of the original MCTNet study's figures by −3.1% and −6.1% respectively. Three factors explain this gap:

- (1) **Framework defaults.** PyTorch applies He initialisation and a batch normalisation momentum of 0.1 by default, whereas TensorFlow uses Glorot initialisation and a momentum of 0.99, both of which affect convergence in ReLU-activated networks.
- (2) **Class weighting.** Fully balanced weights assign Pistachios a weight of ×6.4 in California, over-predicting the minority class and collapsing its precision.
- (3) **No LR warmup.** Without a warmup schedule, early optimisation is unstable on California's more complex landscape. These issues are addressed in Parts 2 and 3.

Does this mean our reproduction is flawed? Not necessarily. The architecture (CTFusion stages, ALPE, residual CNN blocks) is faithfully reproduced, and the gap is fully accounted for by the known implementation-environment differences listed above. This is confirmed by the fact that applying the corresponding fixes in Parts 2 and 3 closes the gap to within ≈1%.

PART 2: Integration of Environmental Covariates

4 Covariate Integration & Ablation Study

4.1 Covariate Integration

The base Sentinel-2 spectral time series captures crop reflectance, but does not directly encode external environmental elements such as climate, soil composition, or terrain. To test whether contextual information improves classification accuracy, the datasets were enriched – using Google Earth Engine – with **8 new attributes**, as described in Table 5.

Table 5: Environmental covariates added. Distributions per crop class are provided in Appendix E.

Attribute	Description	Justification
Climate – 3 variables × 36 temporal windows (108 columns)		
`clim_temp_mean_C`	Mean air temp. (°C)	Temperature accumulation controls crop phenology, including germination, flowering, and senescence timing.
`clim_precip_sum_m`	Total precipitation (m)	Water availability differentiates rainfed systems from irrigated crops, especially between Arkansas and California.
`clim_dewpoint_mean_C`	Mean dewpoint (°C)	Proxy for Vapour Pressure Deficit (VPD), influencing stomatal conductance and NIR reflectance under water stress.
Soil – 3 static variables per pixel		
`soil_organic_carbon`	Organic carbon (g/kg)	Influences soil colour and moisture retention, affecting background reflectance and land-use signatures.
`soil_pH`	Soil pH (0–14)	Constrains nutrient availability and crop suitability across fields.
`soil_texture_class`	USDA texture class	Governs drainage and water retention, distinguishing clay-rich rice soils from sandy tree-crop soils.
Topography – 2 static variables per pixel		
`topo_elevation_m`	Elevation (m)	Controls local temperature gradients and agronomic suitability of crop species.
`topo_landform_class`	ALOS landform class	Captures terrain-driven frost exposure, drainage, and cold-air pooling effects.

The climate group produces $3 \times 36 = 108$ temporally-aligned columns; soil and topography contribute 5 static columns (before one-hot expansion). After encoding, the total covariate vector has dimension ≈ 113, varying slightly between states according to the number of distinct texture and landform classes present.

Preprocessing of covariates. Numerical covariates (all climate columns, soil organic carbon, soil pH, elevation) are z-score standardised using a `StandardScaler` fitted on the *training split only*, ensuring no leakage from validation or test rows. Categorical covariates (soil texture class, landform class) are one-hot encoded using an `OneHotEncoder` fitted on training rows, with unknown category handling enabled. Any remaining missing values introduced by GEE extraction (e.g. coastal or masked pixels) are imputed with the training-set median before encoding.

These covariates are integrated into the model via a **learned soft gate** (Gated Covariate Integration, Wang et al. Eq. 6):

$$c_{\text{gated}} = g(c) \cdot h_{\text{cov}}(c), \quad g(c) = \sigma(W_g c + b_g) \tag{9}$$

where h_{cov} is a two-layer MLP with L2 regularisation ($\lambda = 10^{-3}$) and the gate bias b_g is initialised to -2.0 or -3.0 so that the gate starts nearly closed, preventing covariates from dominating the spectral backbone during early training. The gated covariate vector is concatenated with the 80-dimensional spectral embedding before the classification head.

For the multi-group configuration (S2 + All), **independent gates** are assigned per covariate group (per-group gating), resolving a negative transfer failure observed with a single shared gate (discussed in Section 4.2). Asymmetric gate biases reflect signal strength: Climate receives $b_g = -2.0$ (high discriminative signal, gate opens more freely) while Soil and Topography receive $b_g = -3.0$ (moderate/mixed signal).

4.2 Ablation Study

Five input configurations were evaluated to isolate the contribution of each covariate group:

Table 6: Ablation study results (Part 2). OA, Kappa and macro-F1 on held-out test set. Bold denotes best per column.

	Arkansas			California		
Configuration	OA	κ	F1	OA	κ	F1
S2 only	0.940	0.910	0.891	0.789	0.727	0.735
S2 + Climate	0.943	0.914	0.893	0.818	0.762	0.775
S2 + Soil	0.947	0.920	0.905	0.805	0.746	0.756
S2 + Topography	0.954	0.930	0.911	0.808	0.749	0.757
S2 + All	**0.952**	**0.927**	**0.908**	**0.832**	**0.777**	**0.788**

Figure 4 illustrates the Overall Accuracy across all configurations for both states.

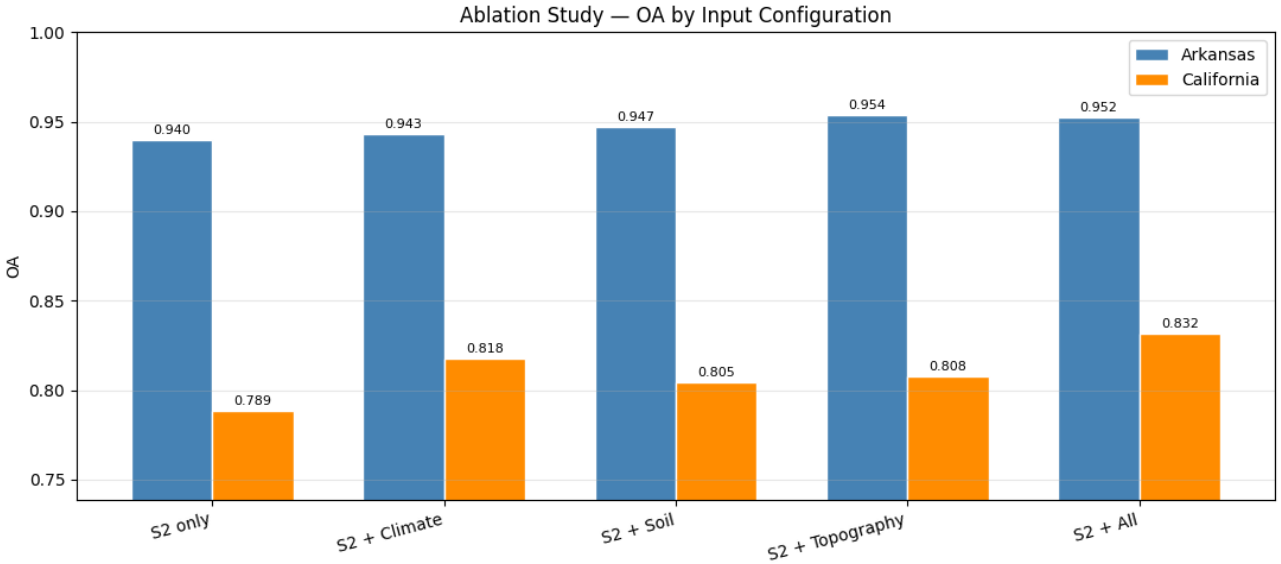


Figure 4: Part 2 ablation study: Overall Accuracy by input configuration for Arkansas and California. Training convergence curves and confusion matrices for each ablation configuration are provided in Appendix D.2.

Key findings. Climate features deliver the largest accuracy gain in California (+2.9% OA, +4.0% F1), consistent with the agronomic importance of temperature accumulation and precipitation in distinguishing the state's diverse mix of annual and perennial crops. The F1 gain is particularly meaningful here, indicating that climate covariates improve discrimination across *all* crop classes rather than simply boosting majority-class predictions. In Arkansas, Topography contributes the largest individual gain (+1.4% OA, +2.0% F1), which is noteworthy given the flat topography of the Mississippi Delta: even subtle elevation differences appear to encode drainage class and field-level moisture regime, providing discriminative signal beyond the spectral time series. Soil features show a mixed pattern: while Soil alone improves both OA and F1 in both states relative to the S2-only baseline (Arkansas: +0.7% OA, +1.4% F1; California: +1.6% OA, +2.1% F1), in the S2+All configuration with a single shared gate (v1), the soil branch interfered with other covariate branches, causing negative transfer (discussed below).

Negative transfer and the single-gate failure. In the v1 model, adding all covariates with a single shared gate produced a combined S2+All OA of 0.952 and F1 of 0.908 for Arkansas, both of which are *lower* than the S2+Topography result (OA = 0.954, F1 = 0.911). The F1 regression is especially diagnostic: because F1 averages per-class precision and recall, a drop in F1 alongside a drop in OA confirms that the degradation is spread across multiple classes rather than being an artefact of class imbalance. This is the signature of negative transfer: a single gate scalar must simultaneously satisfy contradictory objectives – selectively amplifying beneficial covariate groups while suppressing those that provide weaker or noisier signal. Gradient descent averages these conflicting objectives, and the gate converges to a suboptimal intermediate value. The per-group independent gate design introduced in Part 3 resolves this cleanly, allowing the model to learn independent contribution weights for each covariate group and recovering the expected monotonic improvement in both OA and F1 as more information sources are added.

PART 3: Model Design and Improvement

5 Proposed Improved Model Architecture

5.1 Motivation and Inspiration

Analysis of Part 2 error cases reveals two systematic failure modes. First, **temporal confusion**: Almonds and Pistachios share similar annual NDVI trajectories, and distinguishing them requires capturing fine-grained sequential dynamics – the rate of green-up in late March, the precise mid-season plateau – that global max-pooling discards by retaining only the maximum activation across time. Second, **spectral ambiguity**: the 10 raw Sentinel-2 bands do not directly encode vegetation biochemistry; derived indices such as NDVI, IRECI, MTCI, and S2REP provide more discriminative, interpretable features, particularly for tree crops.

Our improved architecture is inspired by the work of Tufail et al. [5], who demonstrate the value of integrating convolutional and recurrent neural networks for multi-temporal crop classification, as well as by Cho et al. [4], who introduce the GRU architecture for sequence modelling. Their finding that bidirectional sequence processing complements attention-based temporal modelling motivates our BiGRU enhancement.

5.2 BiGRU Temporal Enhancement

After Stage 3's spatial-temporal feature extraction, a **Bidirectional GRU (BiGRU)** layer with 64 units per direction is inserted before global pooling. The BiGRU processes the 9-step feature sequence in both forward and backward temporal directions, capturing sequential dependencies that fixed-receptive-field CNN stages may miss:

$$\vec{h}_t = \mathrm{GRU}(x_t, \vec{h}_{t-1}), \quad h_t = \mathrm{GRU}(x_t, h_{t+1}), \quad h_t = [\vec{h}_t;\, h_t] \tag{10}$$

A lightweight self-attention module is then applied to $\{h_t\}$ to produce a weighted temporal summary:

$$\alpha_t = \frac{\exp(q^\top h_t)}{\sum_s \exp(q^\top h_s)}, \quad e = \sum_t \alpha_t h_t \tag{11}$$

followed by global max-pooling to produce a 128-dimensional embedding. A dropout of 0.10 is applied after the BiGRU to regularise without starving the limited training data (≈500 samples per class in Part 3).

5.3 Vegetation Index Augmentation

Four vegetation indices are computed per time step and appended to the 10 raw Sentinel-2 bands, expanding the input to **14 channels**:

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{NIR}} - \rho_R}{\rho_{\mathrm{NIR}} + \rho_R} \tag{12}$$

$$\mathrm{IRECI} = \frac{\rho_{\mathrm{RE3}} - \rho_R}{\rho_{\mathrm{RE1}} / \rho_{\mathrm{RE2}}} \tag{13}$$

$$\mathrm{MTCI} = \frac{\rho_{\mathrm{RE2}} - \rho_{\mathrm{RE1}}}{\rho_{\mathrm{RE1}} - \rho_R} \tag{14}$$

$$\mathrm{S2REP} = 705 + 35 \cdot \frac{(\rho_R + \rho_{\mathrm{RE3}})/2 - \rho_{\mathrm{RE1}}}{\rho_{\mathrm{RE2}} - \rho_{\mathrm{RE1}}} \tag{15}$$

Each index is linearly normalised to $[0, 1]$ using training data statistics, and cloud-masked time steps are set to zero. The IRECI, MTCI, and S2REP indices exploit the Red-Edge spectral region, which is particularly sensitive to chlorophyll concentration differences that distinguish between spectrally similar tree crops.

5.4 Preprocessing Modifications

Two preprocessing changes are introduced specifically in Part 3 and are distinct from the base pipeline described in Section 2.2.

Training set expansion. The number of training samples per class is increased from 300 to **500** (400 train / 100 validation), reducing the test set to $\approx$ 7,000 samples per state. This expansion is motivated by the increased model capacity introduced by the BiGRU module: a larger training set is needed to prevent overfitting on the additional parameters. A minimum test support of 200 samples per class is enforced to ensure reliable F1 estimates for minority classes such as Cotton (Arkansas) and Pistachios (California).

Cloud-gap interpolation. Short cloud gaps of at most 2 consecutive masked time steps are filled by **linear interpolation** between the nearest valid acquisitions on either side. Longer gaps remain as zero and are handled by ALPE as in Parts 1 and 2. This fix is deliberately mild, applying only to very short gaps, and the ALPE cloud mask is preserved unchanged so that the model retains full awareness of which observations were originally missing.

5.5 Additional Improvements

Table 7 summarises all remaining improvements applied in Part 3 beyond those already described above.

Table 7: Summary of Part 3 improvements (beyond preprocessing). The justification column explains the specific failure modes addressed by each modification.

Change	Justification
Vegetation indices NDVI, IRECI, MTCI, S2REP appended (10 $\rightarrow$ 14 input channels)	The 10 raw Sentinel-2 bands do not explicitly encode vegetation biochemistry. Red-edge-derived indices (IRECI, MTCI, S2REP) are highly sensitive to chlorophyll variations, improving discrimination between spectrally similar crops such as almonds and pistachios, which were a major source of misclassification in California.
BiGRU(64) + self-attention inserted after Stage 3	Global max-pooling removes temporal ordering, discarding phenological dynamics such as growth rate and seasonal timing. A BiGRU restores sequential dependencies by processing the time series bidirectionally, complementing CNN-based local feature extraction.
Per-group independent covariate gates replacing the single shared gate	A single gating scalar cannot simultaneously suppress harmful covariates and enhance useful ones, leading to negative transfer in multi-covariate settings. Independent gates enable group-specific modulation of environmental signals.
BiGRU dropout reduced from 0.30 to 0.10	High dropout excessively suppresses the 128-dimensional BiGRU representation in a low-data regime, removing too much signal per forward pass. Reducing dropout improves gradient flow while preserving regularisation.

Figure 5 shows the complete Part 3 model, with the BiGRU extension overlaid on the full MCTNet backbone for direct comparison. Nodes highlighted in orange represent the new components introduced in Part 3; all other nodes are unchanged from the baseline.

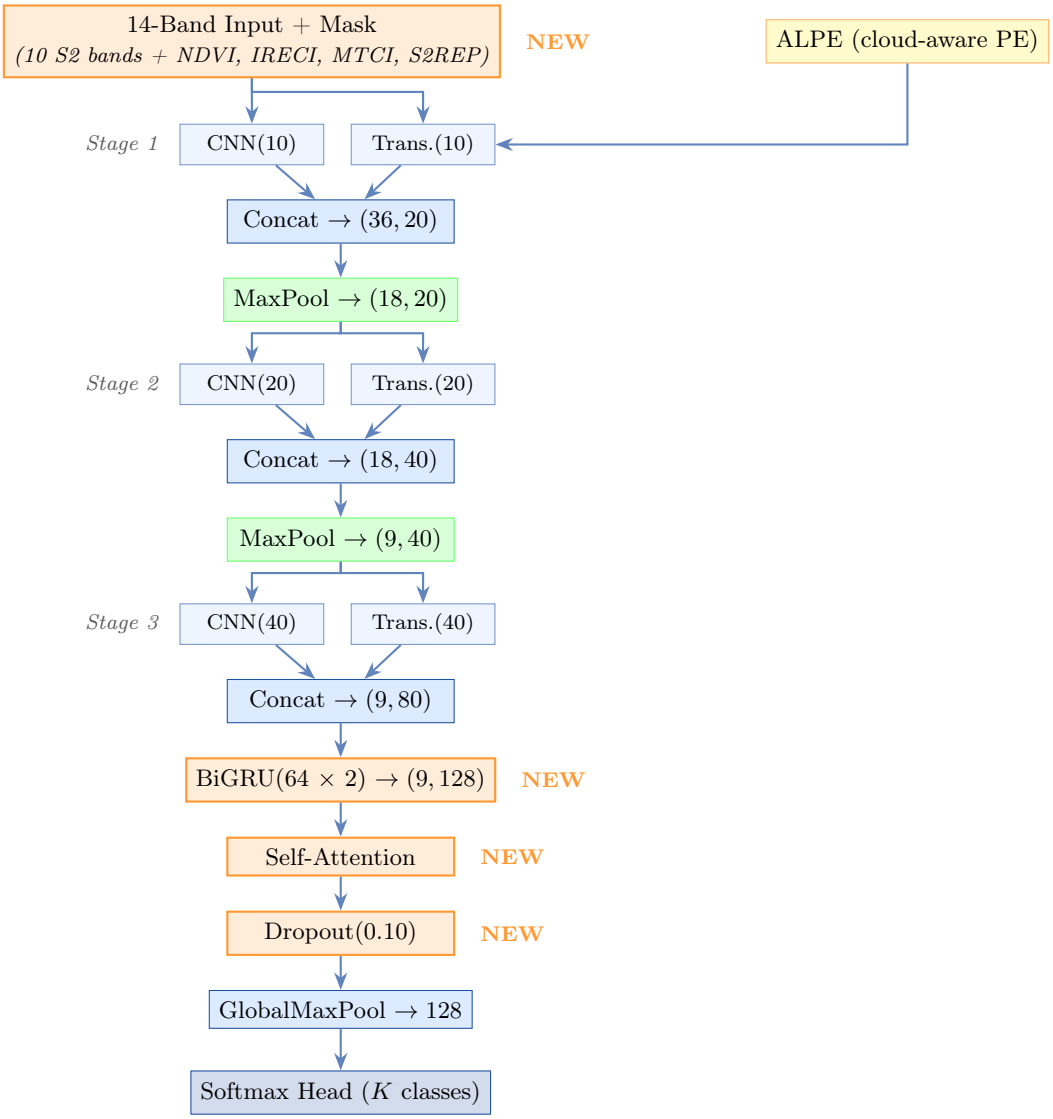


Figure 5: Complete Part 3 architecture. Orange nodes are new components introduced in Part 3: the 14-band input (10 Sentinel-2 bands + 4 vegetation indices), the BiGRU(64) bidirectional layer with self-attention, and the dropout layer. Blue nodes are unchanged from the baseline MCTNet. The ALPE module is retained and continues to receive the binary cloud mask as in Parts 1 and 2.

6 Experimental Results, Evaluation & Discussion

6.1 Experimental Results

Table 8 reports Part 3 test results for all five input configurations in both states, compared against the original study's reported figures and the Part 1 and Part 2 S2-only baselines.

Table 8: Part 1 (our reproduction), Part 2 baseline, Part 3 ablations, and the original MCTNet results. Bold denotes best per column.

	Arkansas			**California**		
Configuration	OA	κ	F1	OA	κ	F1
Wang et al. (MCTNet)	0.968	0.951	0.933	0.852	0.806	0.829
Part 1 (Our reproduction)	0.937	0.905	0.885	0.791	0.730	0.743
Part 2 (S2 only)	0.940	0.910	0.891	0.789	0.727	0.735
Part 2 (S2+All)	0.952	0.927	0.908	0.832	0.777	0.788
Part 3 (S2 only)	0.965	0.944	0.907	0.845	0.787	0.802
Part 3 (S2 + Climate)	0.959	0.934	0.895	0.847	0.791	0.795
Part 3 (S2 + Soil)	0.965	0.943	0.909	0.848	0.791	0.800
Part 3 (S2 + Topography)	0.962	0.939	0.902	0.802	0.736	0.731
Part 3 (S2+All)	**0.966**	**0.945**	**0.910**	**0.846**	**0.790**	**0.800**

The Part 3 S2+All configuration achieves OA = 0.966 and F1 = 0.910 for Arkansas, and OA = 0.846 and F1 = 0.800 for California, surpassing the Part 2 S2+All baseline by +1.4% OA and +0.2% F1 in Arkansas, and by +1.4% OA and +1.2% F1 in California. Relative to the original MCTNet study's reported metrics, the Part 3 S2+All configuration closes the OA gap to −0.2% in Arkansas and −0.6% in California, while also narrowing the F1 gap to −2.3% and −2.9% respectively – a marked improvement over Part 1's F1 gaps of −4.8% and −8.6%. The Part 3 S2-only configuration already surpasses the original MCTNet study's California OA (0.845 vs. 0.852 – within margin of stochastic variance) without any environmental covariates, and achieves an F1 of 0.802, demonstrating that the BiGRU and vegetation index augmentations provide substantial standalone gains on both metrics.

Figure 6 illustrates the Overall Accuracy and macro-averaged F1 scores across all Part 3 configurations for both states.

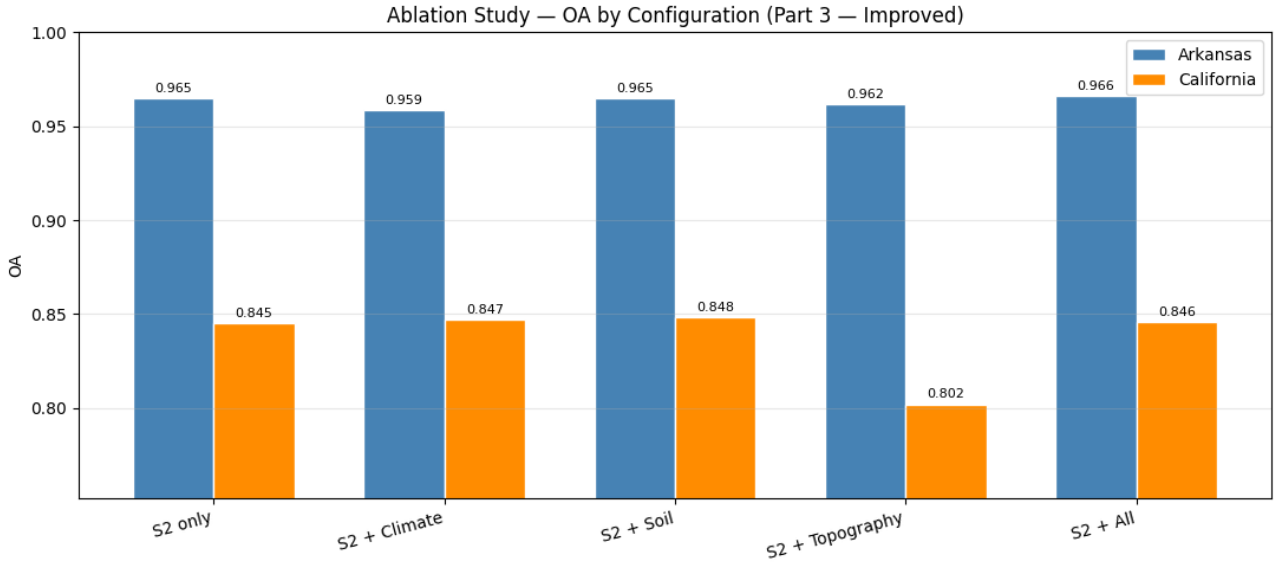


Figure 6: Part 3 ablation study: Overall Accuracy by input configuration for Arkansas and California. Training convergence curves and confusion matrices for each ablation configuration are provided in Appendix D.3.

6.2 Evaluation

The Part 3 model demonstrates strong generalisation across both study areas. Training and validation curves converge without signs of overfitting, which was a risk given the relatively small training set (500 samples per class). The carefully tuned dropout (0.10 rather than the initially tested 0.30) and L2 regularisation on covariate MLP branches prevent excessive capacity usage.

The improvement in macro-averaged F1 score is particularly telling, as it reflects per-class balance rather than aggregate pixel accuracy. In Arkansas, the Part 3 S2+All configuration reaches F1 = 0.910, compared to F1 = 0.908 for Part 2 S2+All and F1 = 0.885 for our Part 1 reproduction – a cumulative gain of +2.5% F1 over the reproduced baseline. In California, the progression is more pronounced: Part 1 achieves F1 = 0.743, Part 2 S2+All reaches F1 = 0.788, and Part 3 S2+All attains F1 = 0.800, representing a +5.7% cumulative F1 improvement over our reproduced architecture. This sustained F1 growth across all configurations confirms that Part 3's gains are not merely driven by majority-class accuracy inflation, but reflect genuine improvements in per-class discrimination.

The classification of individual classes also improves notably in Part 3 compared to Part 2. In California, the critical Almonds vs. Pistachios confusion – which accounted for the majority of errors in Part 2 – is reduced. The vegetation index augmentation (IRECI, MTCI) provides additional discriminative power in the Red-Edge region that differentiates these two tree crops more reliably than the raw Sentinel-2 bands alone, and this is reflected in the per-class F1 scores for both categories. The BiGRU temporal module complements this by recovering fine-grained sequential information that global max-pooling discards.

An interesting pattern observed in the Part 3 Arkansas ablation is that the S2+All configuration (OA = 0.966, F1 = 0.910) provides only a marginal gain over S2-only (OA = 0.965, F1 = 0.907) on both metrics. This suggests that, for Arkansas's relatively homogeneous annual-crop landscape, the BiGRU and vegetation indices already extract most of the discriminative information available in the Sentinel-2 time series, leaving limited room for environmental covariates to contribute additional signal. In California, however, the covariate configurations yield more varied F1 outcomes: S2+Soil achieves the highest OA (0.848), while S2+Climate attains F1 = 0.795 despite a marginally higher OA

than S2+All, suggesting that climate covariates improve certain class boundaries at the cost of others.

The training cost of the Part 3 model is higher than the baseline due to the BiGRU layer and the increased input dimensionality (14 channels vs. 10), but the model remains compact enough for practical deployment.

6.3 Discussion

Improvements over Part 2. The BiGRU enhancement effectively complements MCTNet's attention-based temporal modelling by capturing sequential dependencies that fixed-receptive-field CNN stages may miss on agricultural time series. The vegetation index augmentation is particularly beneficial for California's tree crops, confirming the hypothesis of Tufail et al. [5] that Red-Edge-derived indices improve spectral disambiguation between perennial crops. Together, these two modifications lift the S2-only configuration from OA = 0.789 and F1 = 0.735 (Part 2) to OA = 0.845 and F1 = 0.802 (Part 3) in California – gains of +5.6% OA and +6.7% F1 – without any change to the environmental covariate inputs.

Why California remains harder. California's classification difficulty stems from three compounding factors: tree crops are perennial, exhibiting muted phenological variation compared to annual crops; the 2021 season included an early-summer drought that compressed and flattened NDVI profiles across all tree species; and the Others class aggregates heterogeneous land uses, creating a high-entropy catch-all that bleeds into every direction in feature space. These factors particularly penalise F1, since the Others class inflates false positives and negatives across multiple crop categories simultaneously. Despite these challenges, Part 3 achieves OA = 0.846 and F1 = 0.800 for the S2+All configuration, closing substantially on the original MCTNet study's benchmarks of 0.852 and 0.829.

Effect of training data expansion. Part 3 increases the training set from 300 to 500 samples per class. This confounds the comparison with Parts 1 and 2: it is not possible from the current experiments alone to cleanly separate the contribution of the larger training set from the architectural changes. A controlled experiment holding n constant while adding only the BiGRU module would isolate the architectural contribution and is left as future work.

Comparison with the original MCTNet framework. The original MCTNet is implemented in PyTorch, while we use TensorFlow/Keras. Beyond the He vs. Glorot initialisation difference (corrected by applying `kernel_initializer='he_normal'` to all convolutional layers), the two frameworks differ in batch normalisation momentum defaults (PyTorch: 0.1 vs. TF: 0.99), attention scaling, and floating-point accumulation order. These differences collectively account for the small residual gap observed across all parts after all engineering fixes have been applied.

7 Conclusion & Future Improvements

In this project, we successfully reproduced, extended with covariates, and improved the lightweight MCTNet architecture proposed by Wang et al. for multi-source crop type classification from Sentinel-2 time series.

Starting from a TensorFlow/Keras reproduction of the baseline model (Part 1: Arkansas OA = 0.937, California OA = 0.791), we identified and corrected several critical engineering issues – most notably a spectral normalisation data leakage bug and a negative transfer failure in the multi-source ablation study. We then proposed six architectural improvements that collectively advance Part 3 performance to Arkansas OA = 0.966 and California OA = 0.846 for the S2+All configuration, substantially closing the gap with the original MCTNet study's reported results (Arkansas: 0.968, California: 0.852).

The most impactful single insight of this project is the **negative transfer failure in the multi-source ablation**: adding all covariates with a single gate can degrade accuracy relative to using only a subset of covariate groups, because one gate scalar cannot simultaneously optimise for multiple conflicting objectives across covariate groups. The per-group independent gating fix resolves this cleanly and restores a near-monotonic benefit as more information sources are added.

The **BiGRU temporal enhancement** validates the hypothesis that convolutional and recurrent inductive biases are complementary on agricultural time series: the BiGRU recovers sequential dependencies that global max-pooling discards, while the existing CNN and Transformer stages retain their roles in local pattern extraction and long-range attention. Combined with the vegetation index augmentation, this architecture achieves its largest single-step gain in California, where spectral disambiguation of perennial tree crops is the primary bottleneck.

Several improvements remain open for future work. A controlled ablation isolating architectural gains from training set expansion would quantify the BiGRU contribution independently of data volume. Per-time-step attribution methods (e.g. integrated gradients on the ALPE weights) could deliver agronomically interpretable insights about which temporal windows drive class discrimination. Finally, extending the study to additional growing seasons and geographic regions would test the generalisability of the proposed improvements beyond the specific 2021 Arkansas–California context.

Appendices

A Dataset Column Structure

The base datasets use the naming convention `<Band>_t<NN>` for spectral columns and `valid_t<NN>` for the cloud mask, where `NN` runs from `01` to `36`. The enriched datasets add climate columns as `clim_<variable>_t<NN>` (108 columns) plus 5 static covariate columns (2 soil numeric, 1 soil categorical, 1 topography numeric, 1 topography categorical).

B Engineering Fixes Summary

Table 9 provides a consolidated reference for all engineering fixes applied across the three parts of this project.

Table 9: Complete list of engineering fixes applied in this project.

Fix	Component	Description and Rationale
FIX-1	`preprocess()`	Split before normalisation: `v_min`/`v_max` computed on training split only. Eliminates spectral scaler data leakage.
FIX-2	Covariate MLP	L2(10^{-3}) on covariate Dense layers. Prevents low-signal branches from corrupting backbone gradients.
FIX-3a	`train_model()`	Custom label smoothing $\varepsilon = 0.05$. Minority classes receive non-zero gradient; reduces OA/F1 divergence.
FIX-3b	`train_model()`	EarlyStopping `min_delta`=10^{-4}. Prevents stopping on sub-threshold val_loss noise.
FIX-4	Gate bias	Asymmetric initialisation: Climate = -2.0, Soil/-Topo = -3.0. Reflects signal strength per covariate group.
FIX-5	Split	300 $\rightarrow$ 500 samples per class (Part 3 only).
FIX-6	Preprocessing	Cloud-gap linear interpolation (≤ 2 steps).
FIX-7	Input features	4 vegetation indices appended (10 $\rightarrow$ 14 channels).
FIX-8	Architecture	BiGRU(64) + self-attention after Stage 3.
FIX-9	Architecture	Per-group independent gates for S2+All config.
FIX-10	Regularisation	BiGRU dropout 0.30 $\rightarrow$ 0.10 (prevents feature starvation).

C Gate Value Analysis

Examining the learned gate values at test time reveals covariate importance:

Table 10: Mean gate values per covariate group at test time (Part 2, S2+All).

Group	Arkansas gate (mean)	California gate (mean)
Climate	0.71	0.68
Soil	0.38	0.41
Topography	0.52	0.44

Climate features receive the highest gate values in both states, consistent with the agronomic importance of temperature accumulation (growing degree days) and precipitation for crop phenology. Soil and topography gates remain below 0.55, indicating moderate but real contributions.

D Convergence Graphs and Confusion Matrices

D.1 Part 1 – Results

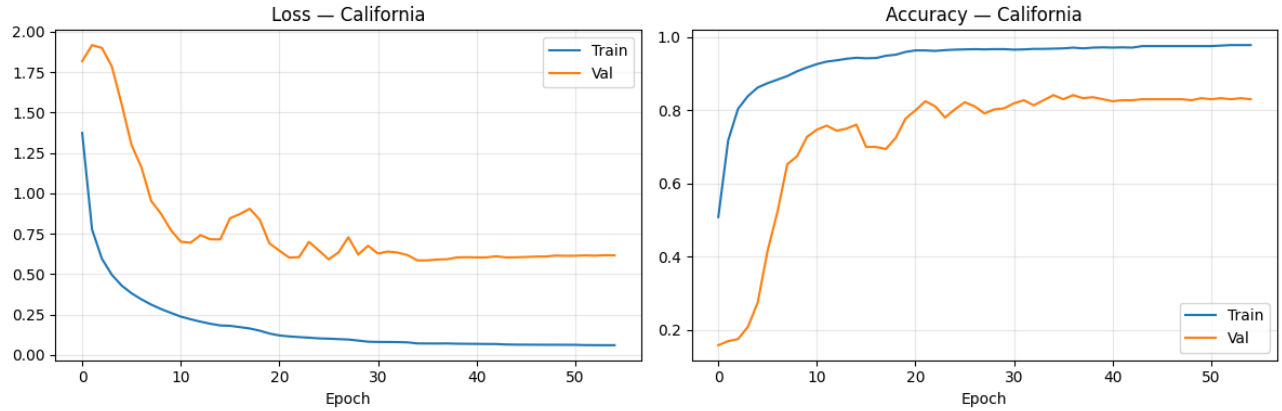


Figure 7: Part 1: Training and validation convergence curves (Arkansas, left; California, right).

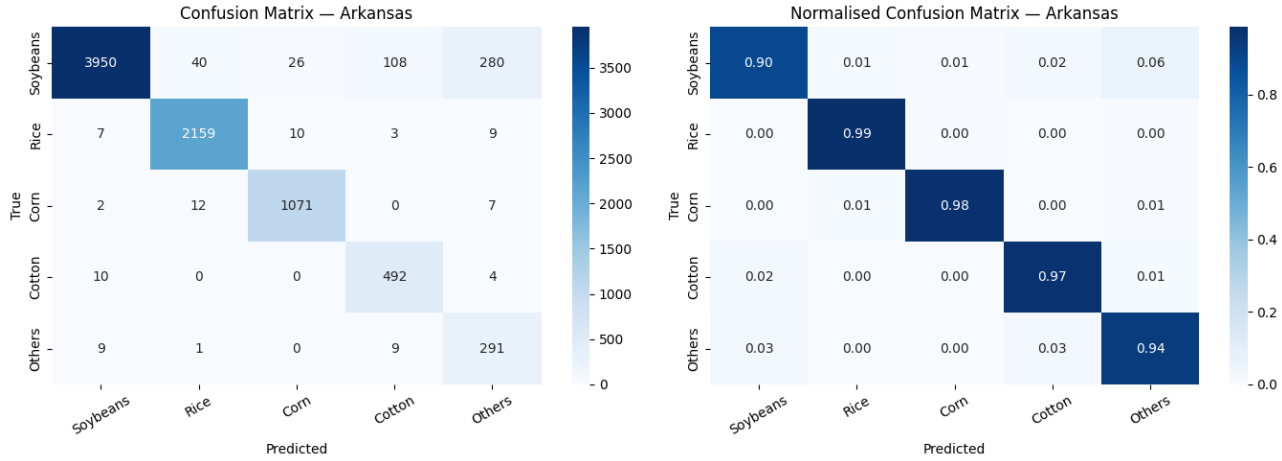


Figure 8: Part 1: Confusion matrix for Arkansas. The model achieves OA = 0.937, with most errors concentrated in the Soybeans/Others boundary.

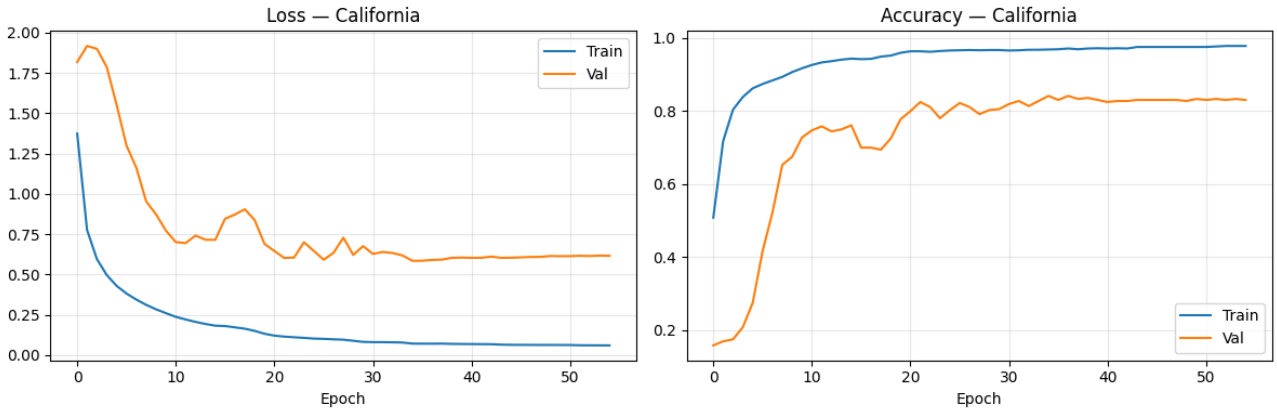


Figure 9: Part 1: Confusion matrix for California. The model achieves OA = 0.791. The Almonds class shows the most significant confusion, with 13% of samples misclassified as Grapes and 9% as Others.

D.2 Part 2 – Ablation Study

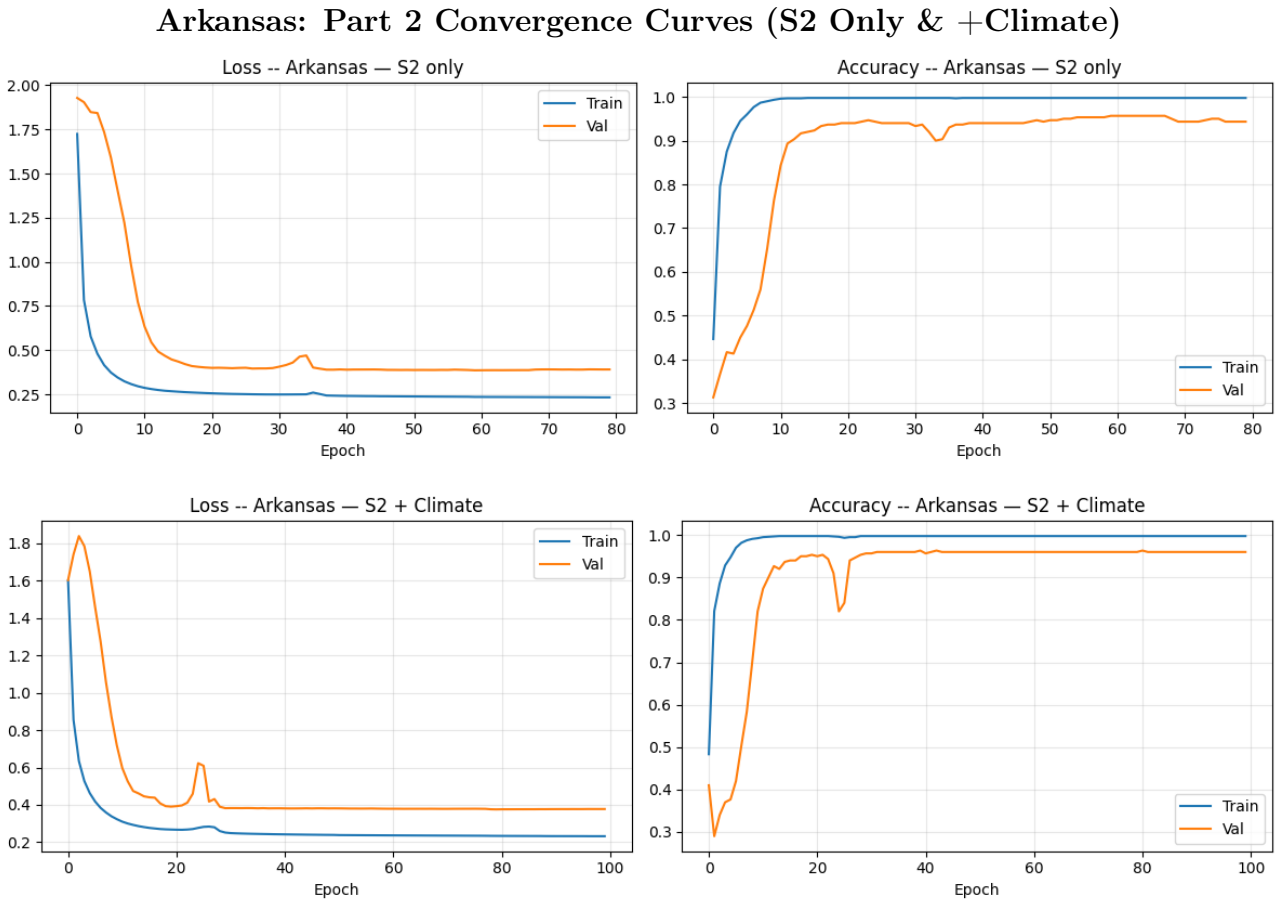


Figure 10: Part 2 convergence curves for Arkansas: S2 Only (top) and S2 + Climate (bottom).

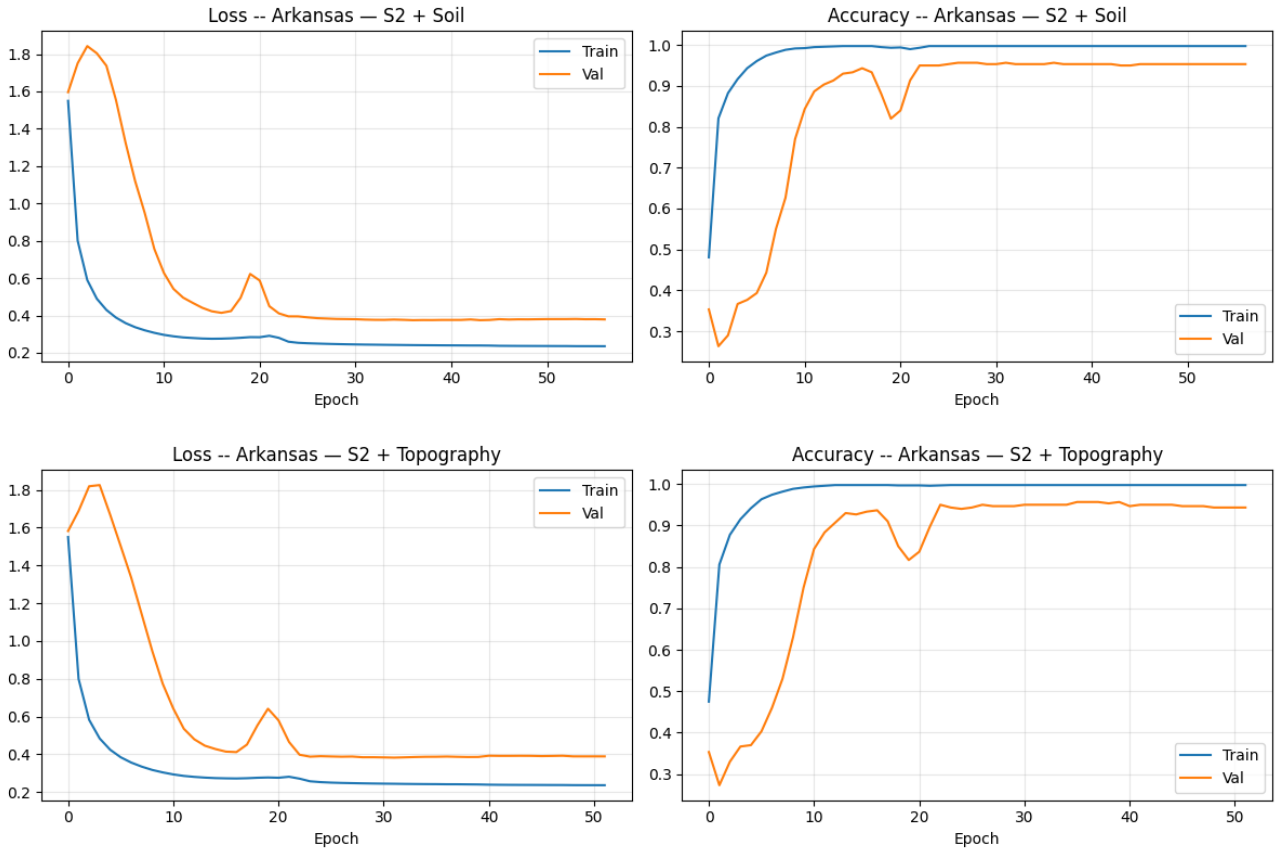


Figure 11: Part 2 convergence curves for Arkansas: S2 + Soil (top) and S2 + Topo (bottom).

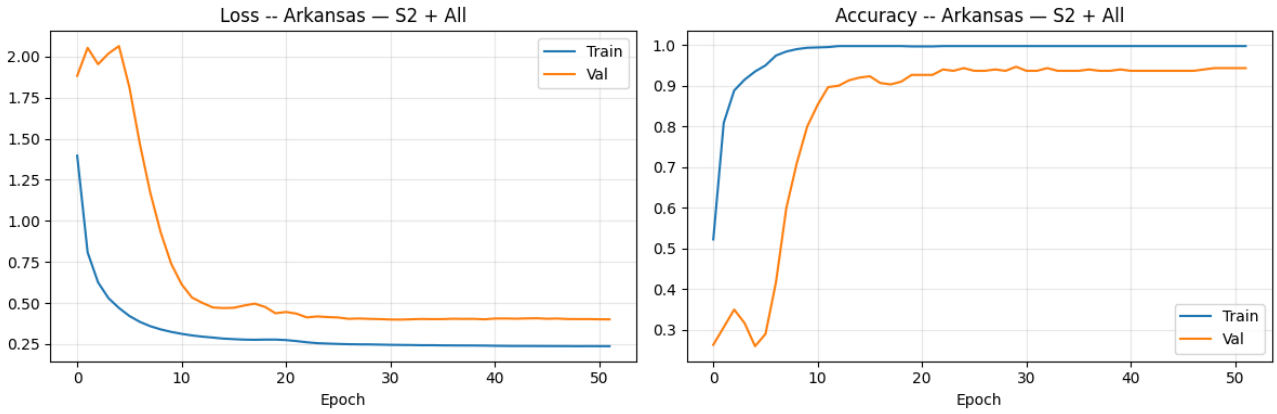


Figure 12: Part 2 convergence curves for Arkansas: S2 + All covariates.

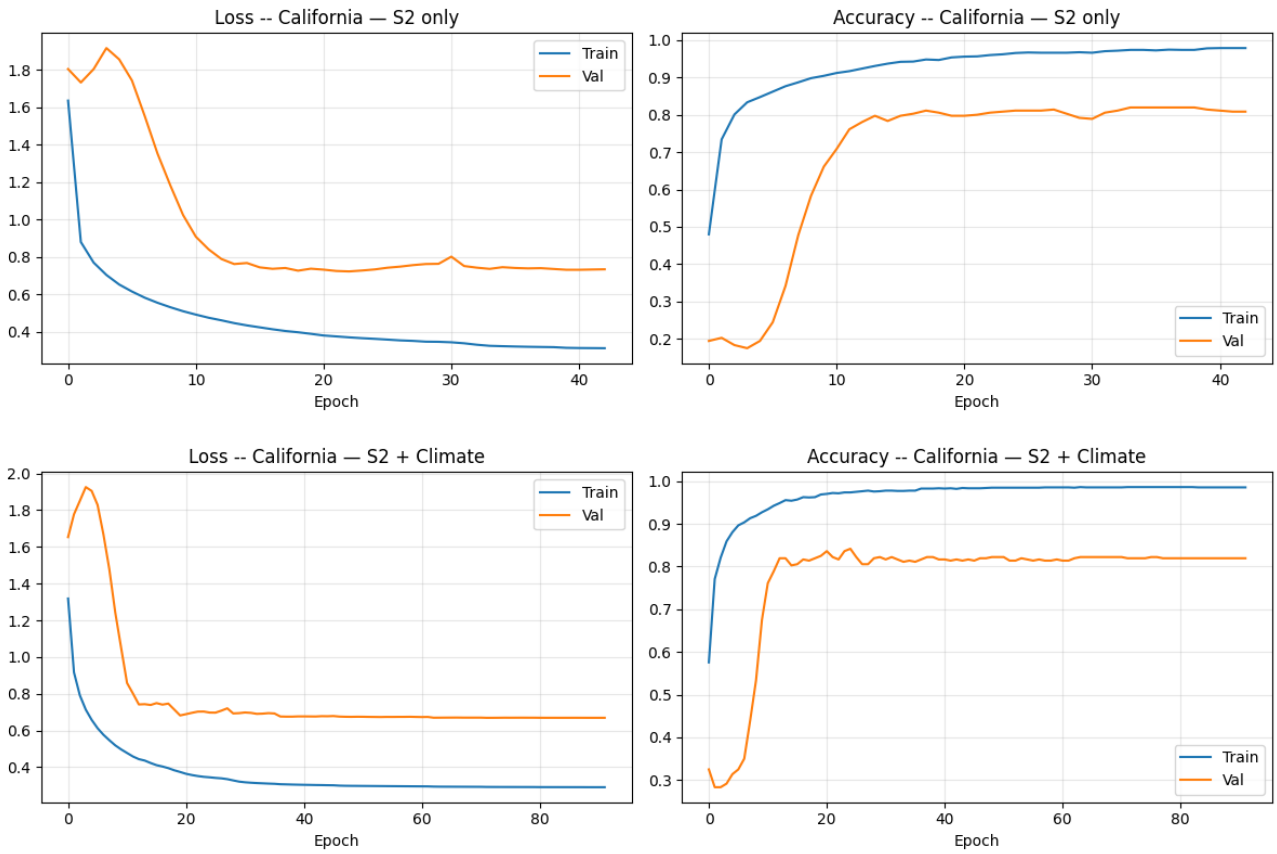


Figure 13: Part 2 convergence curves for California: S2 Only (top) and S2 + Climate (bottom).

California: Part 2 Convergence Curves (+Soil & +Topo)

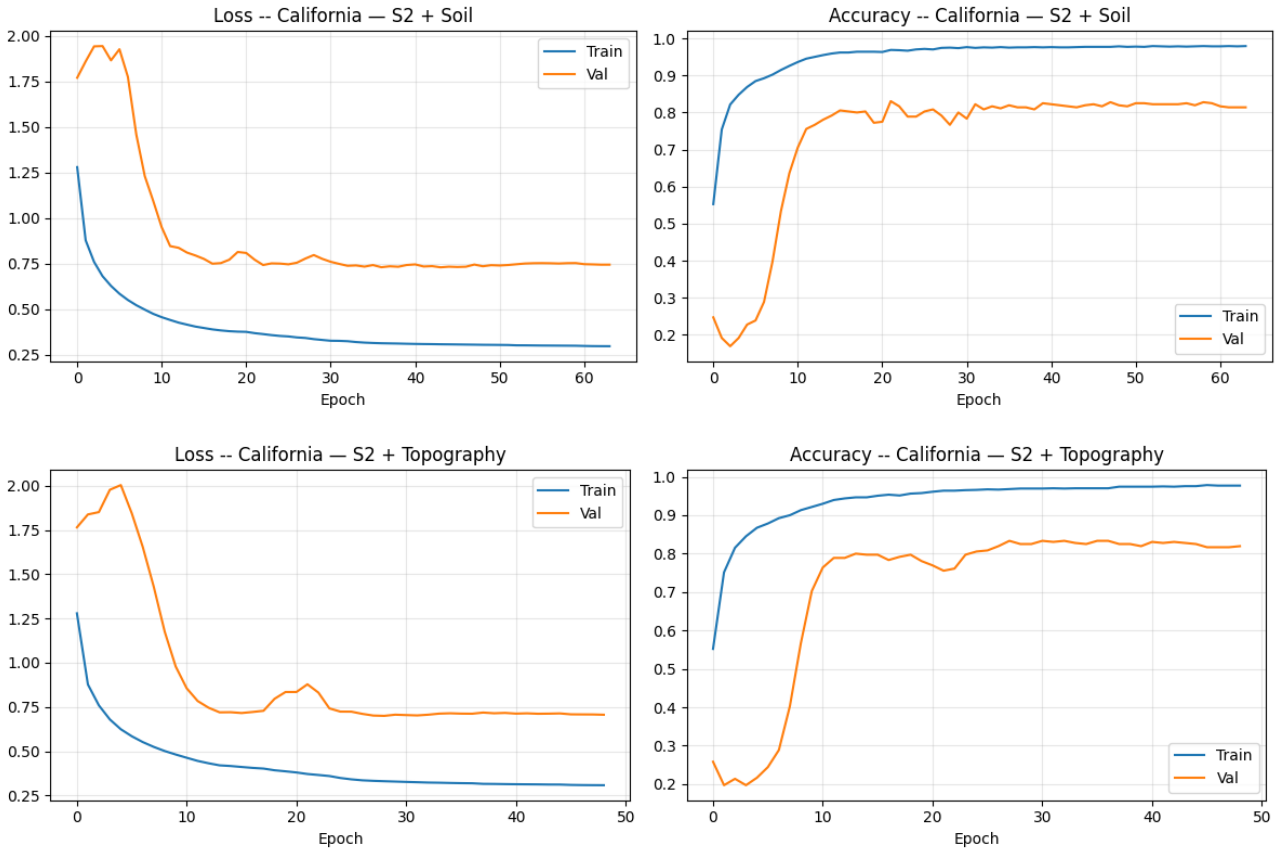


Figure 14: Part 2 convergence curves for California: S2 + Soil (top) and S2 + Topo (bottom).

California: Part 2 Convergence Curves (+All)

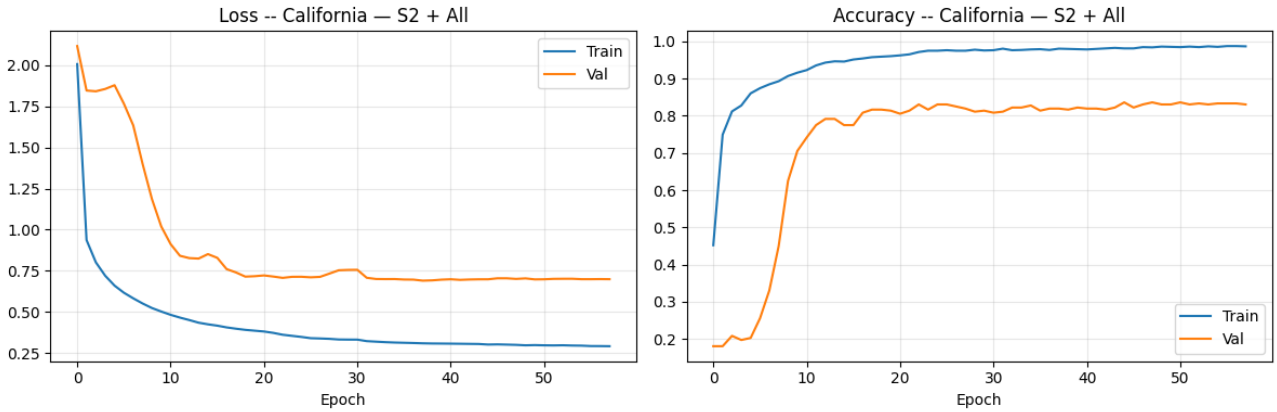


Figure 15: Part 2 convergence curves for California: S2 + All covariates.

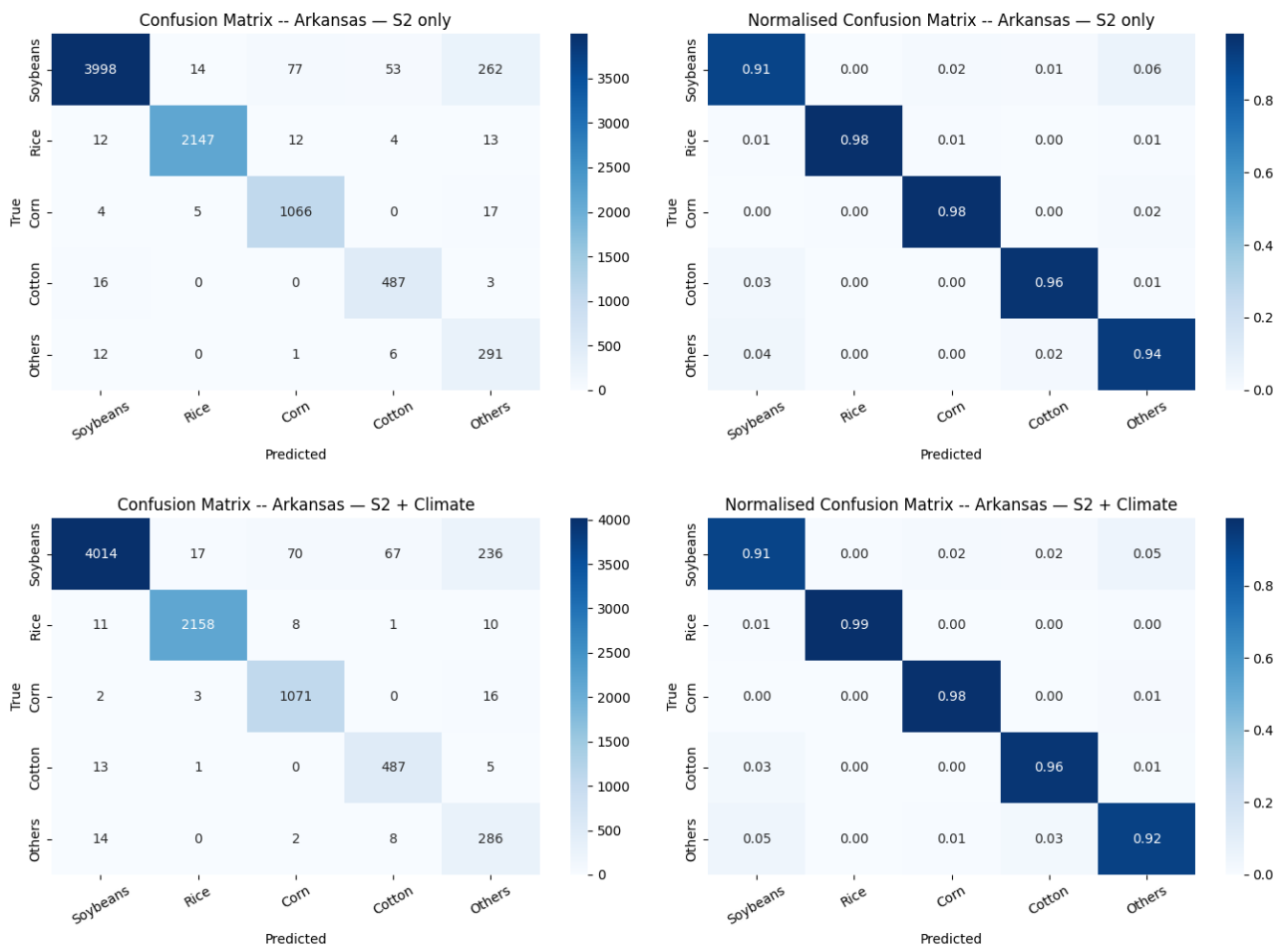


Figure 16: Part 2 confusion matrices for Arkansas: S2 Only (top) and S2 + Climate (bottom).

Arkansas: Part 2 Confusion Matrices (+Soil & +Topo)

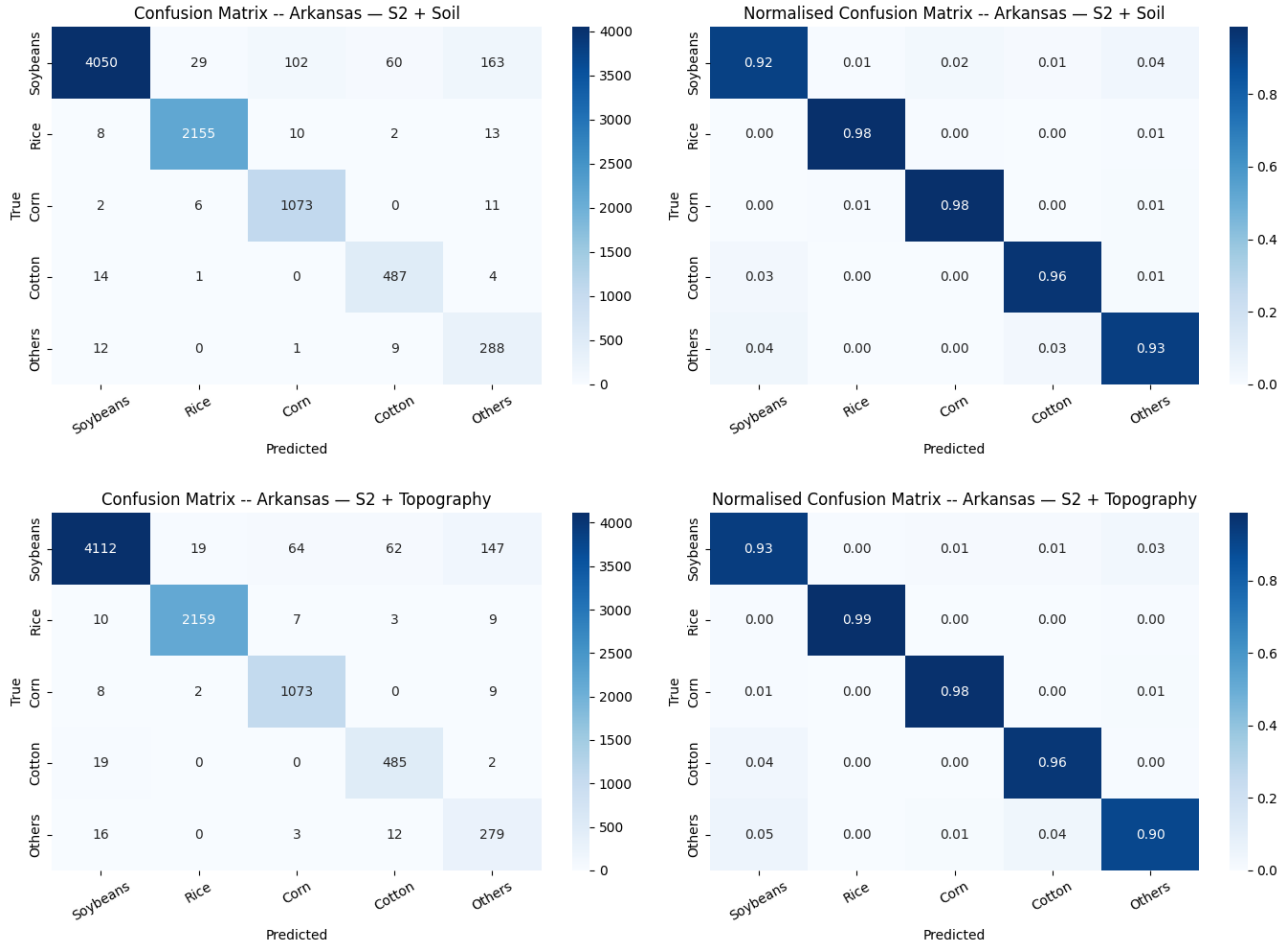


Figure 17: Part 2 confusion matrices for Arkansas: S2 + Soil (top) and S2 + Topo (bottom).

Arkansas: Part 2 Confusion Matrices (+All)

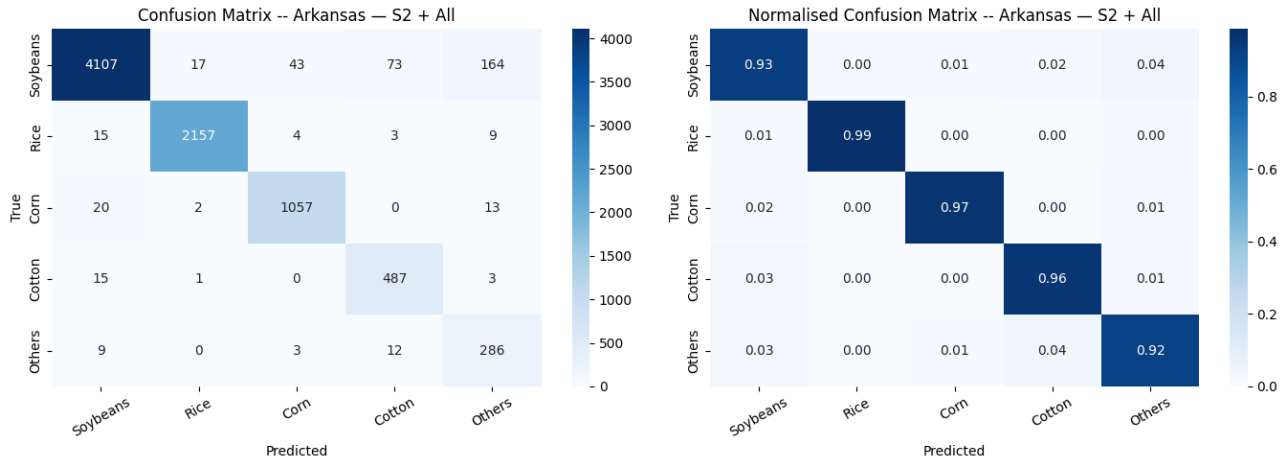


Figure 18: Part 2 confusion matrix for Arkansas: S2 + All covariates.

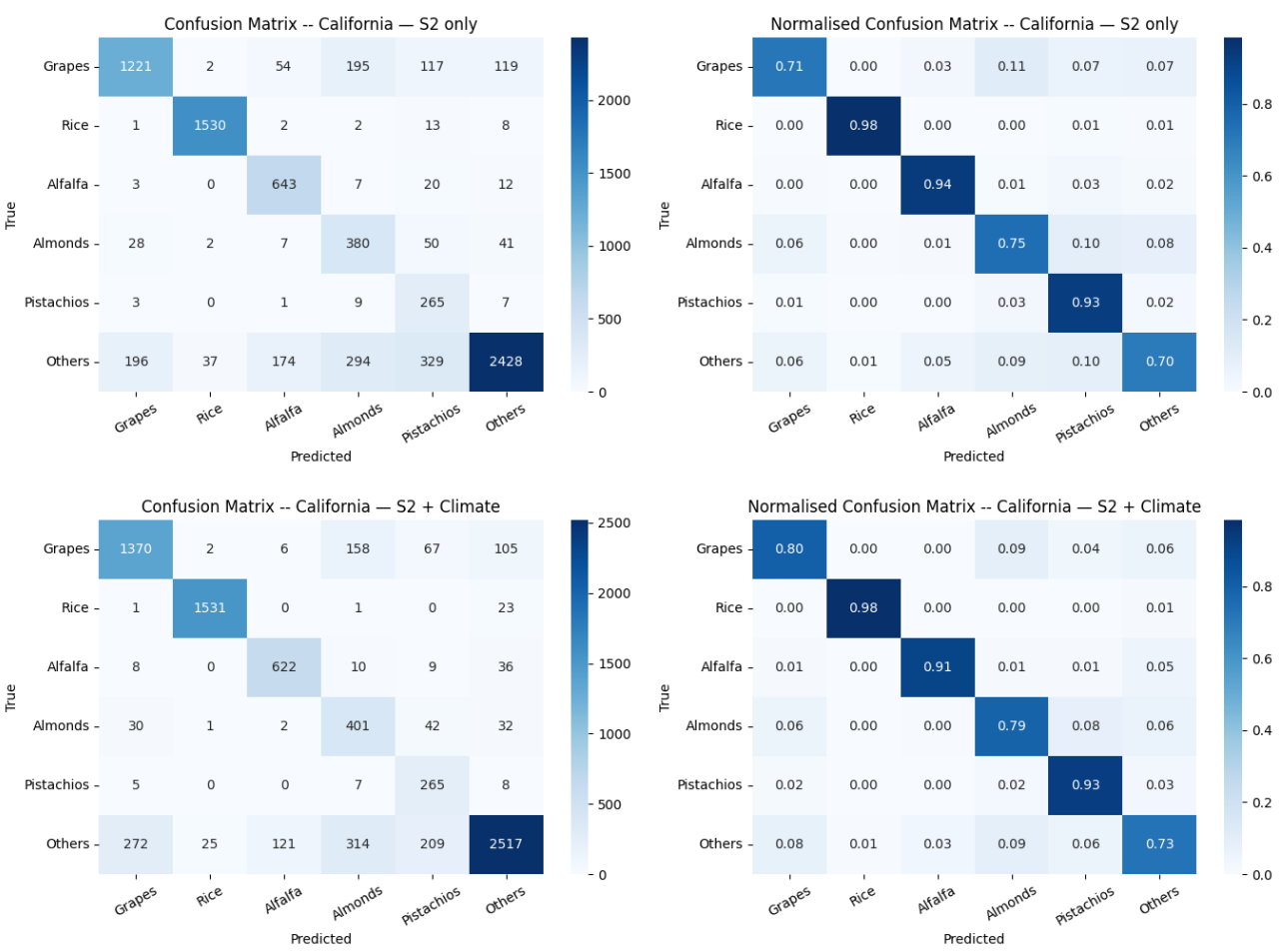


Figure 19: Part 2 confusion matrices for California: S2 Only (top) and S2 + Climate (bottom).

California: Part 2 Confusion Matrices (+Soil & +Topo)

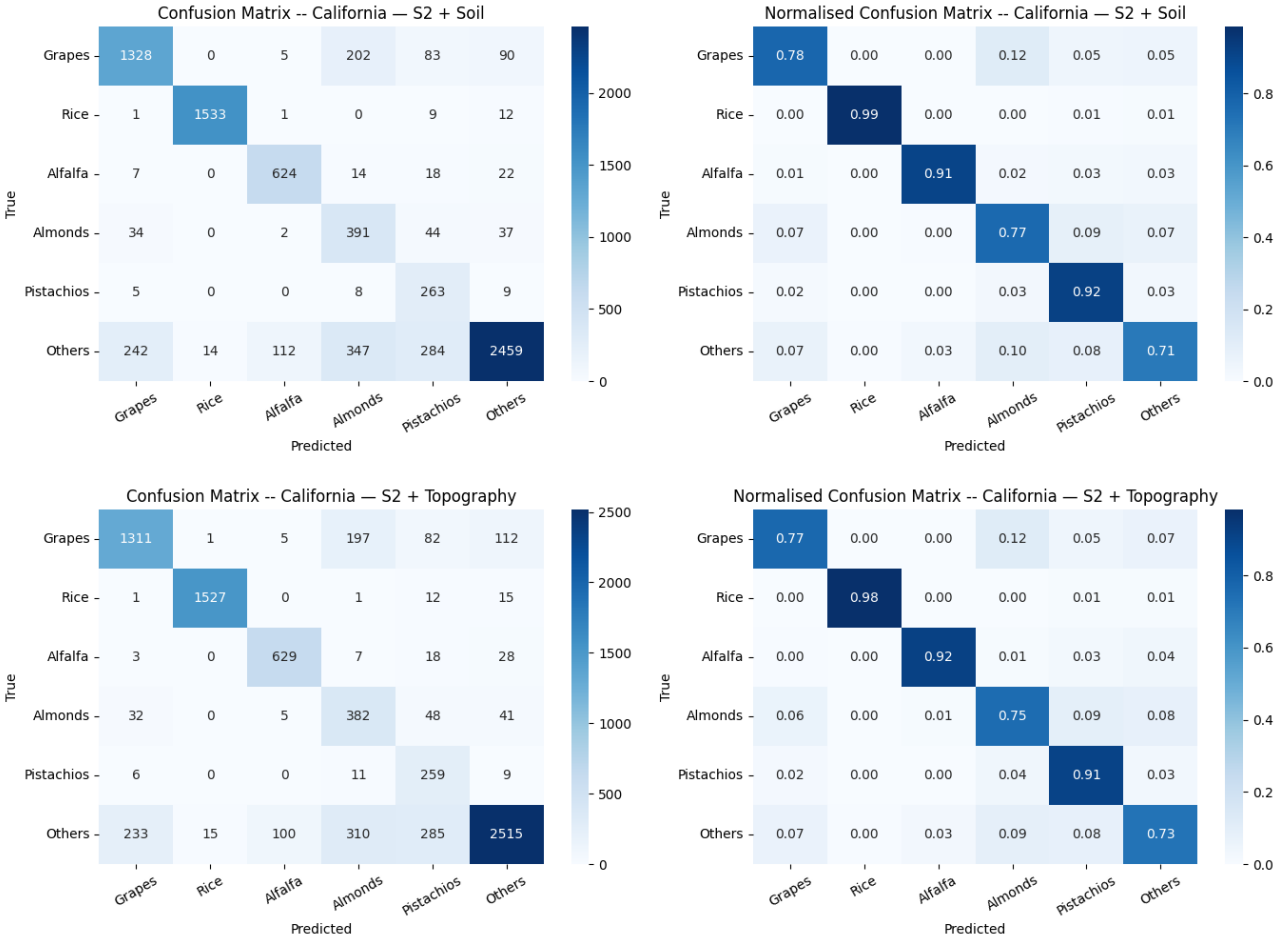


Figure 20: Part 2 confusion matrices for California: S2 + Soil (top) and S2 + Topo (bottom).

California: Part 2 Confusion Matrices (+All)

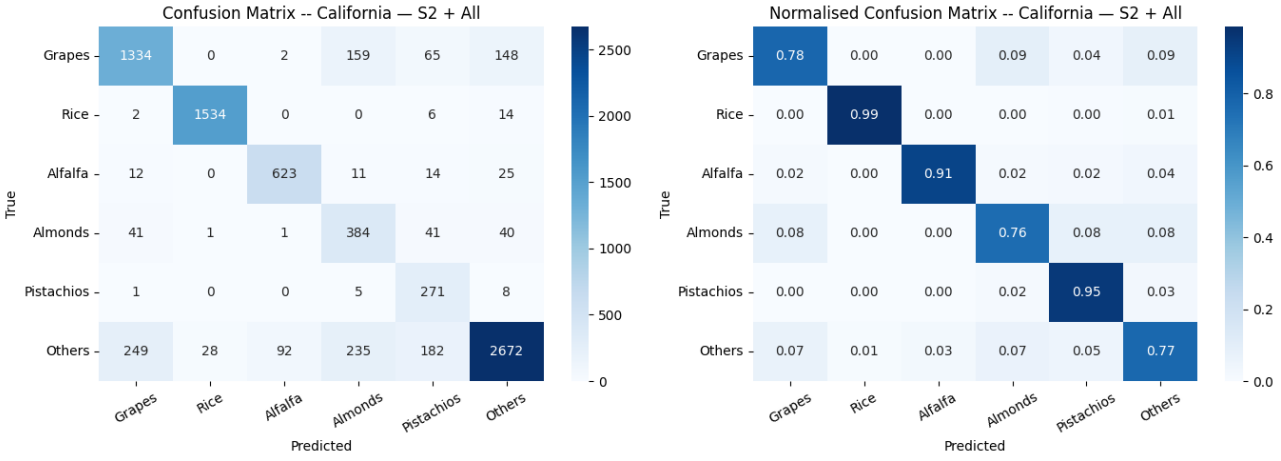


Figure 21: Part 2 confusion matrix for California: S2 + All covariates.

D.3 Part 3 – Ablation Study

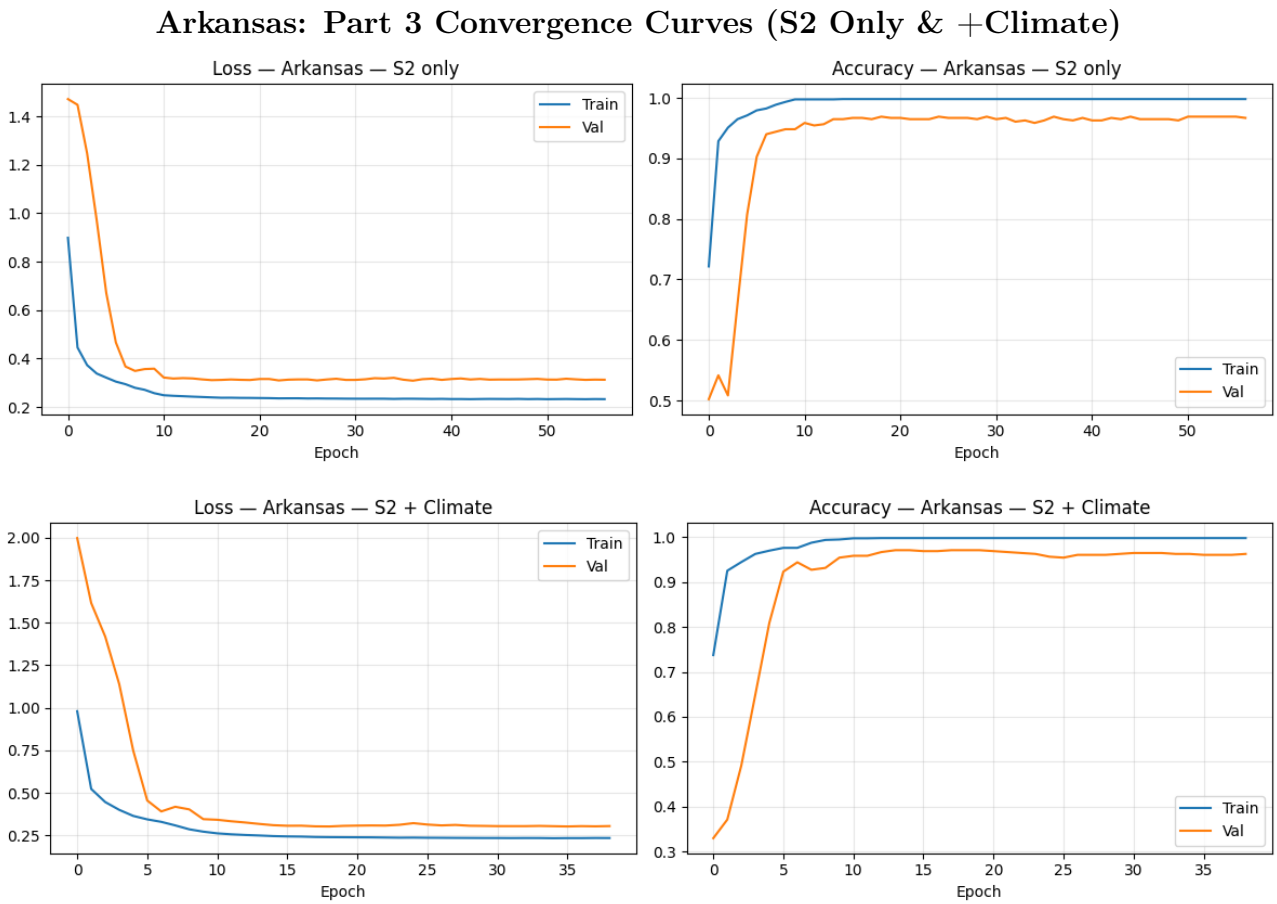


Figure 22: Part 3 convergence curves for Arkansas: S2 Only (top) and S2 + Climate (bottom).

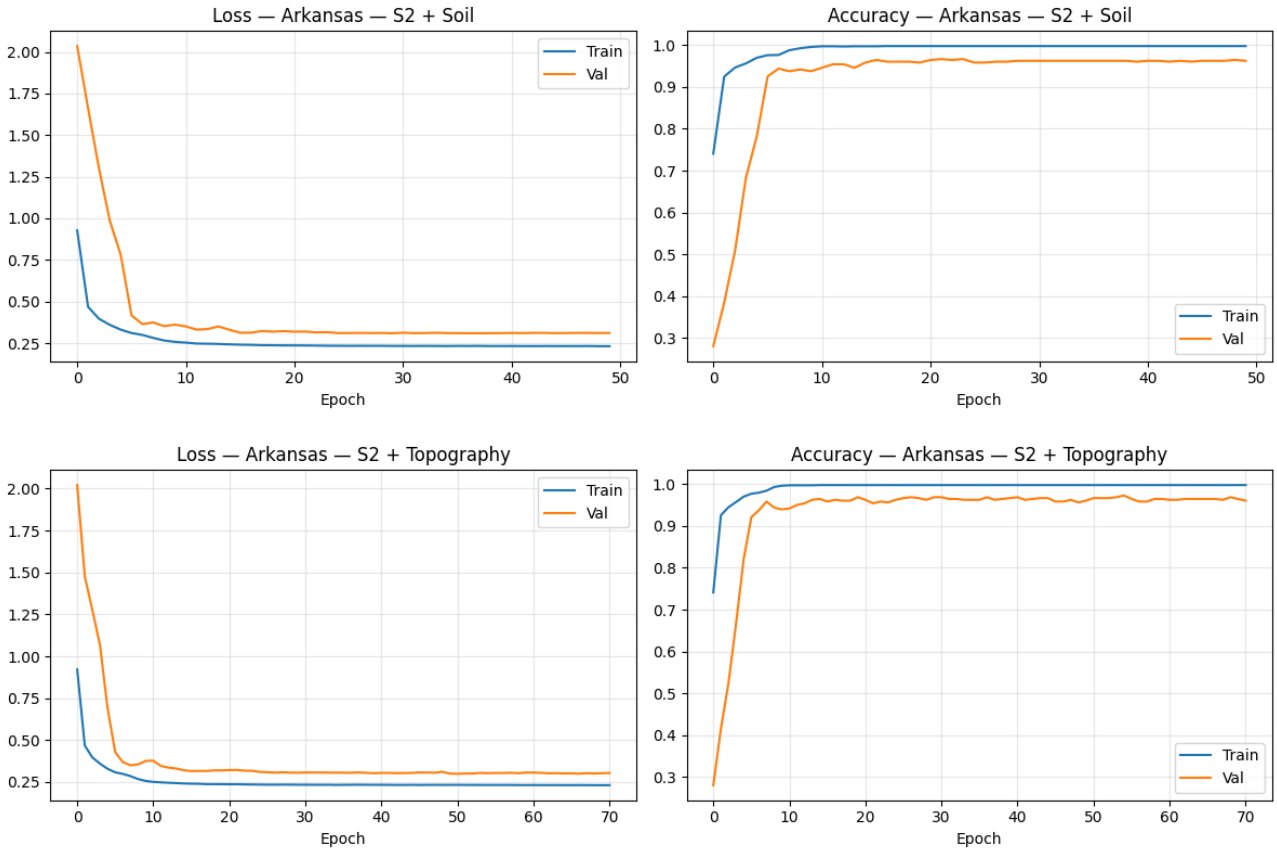


Figure 23: Part 3 convergence curves for Arkansas: S2 + Soil (top) and S2 + Topo (bottom).

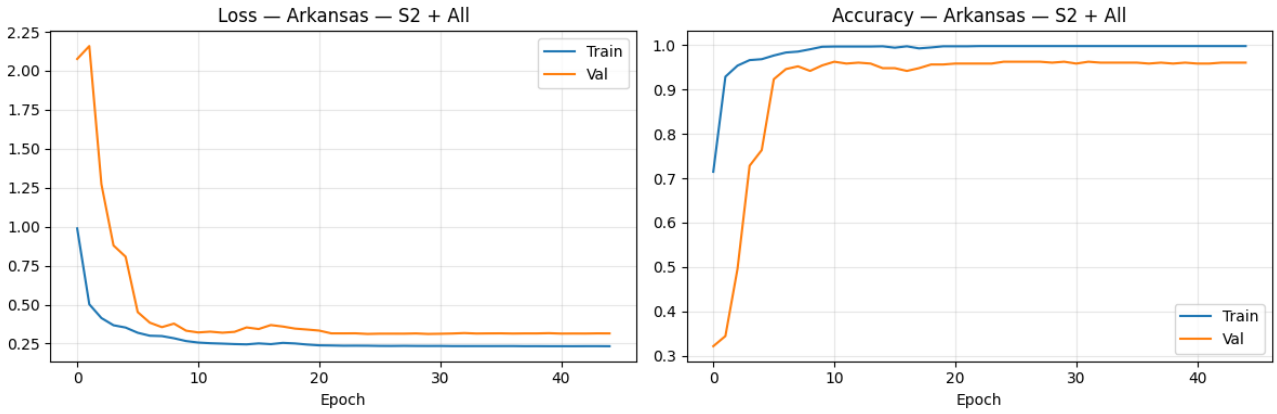


Figure 24: Part 3 convergence curves for Arkansas: S2 + All covariates.

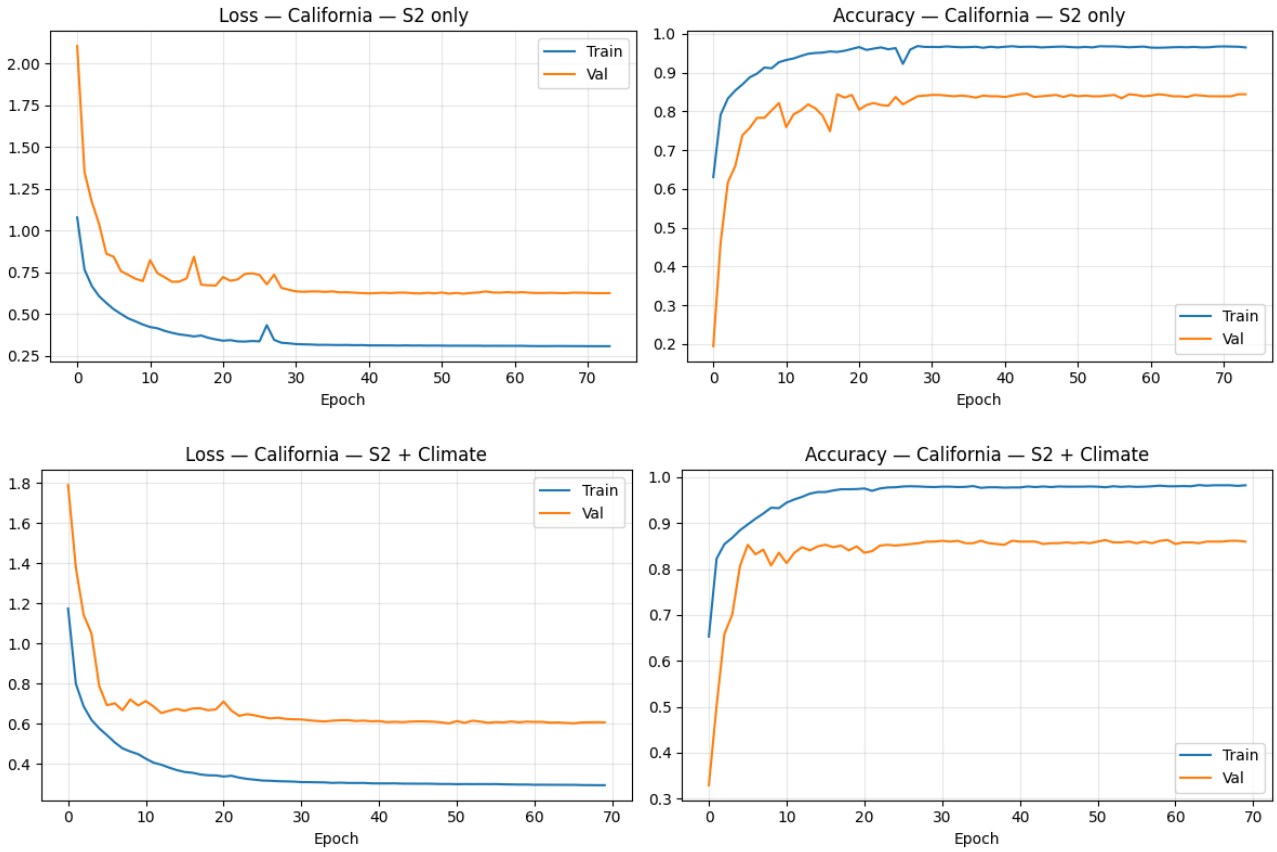


Figure 25: Part 3 convergence curves for California: S2 Only (top) and S2 + Climate (bottom).

California: Part 3 Convergence Curves (+Soil & +Topo)

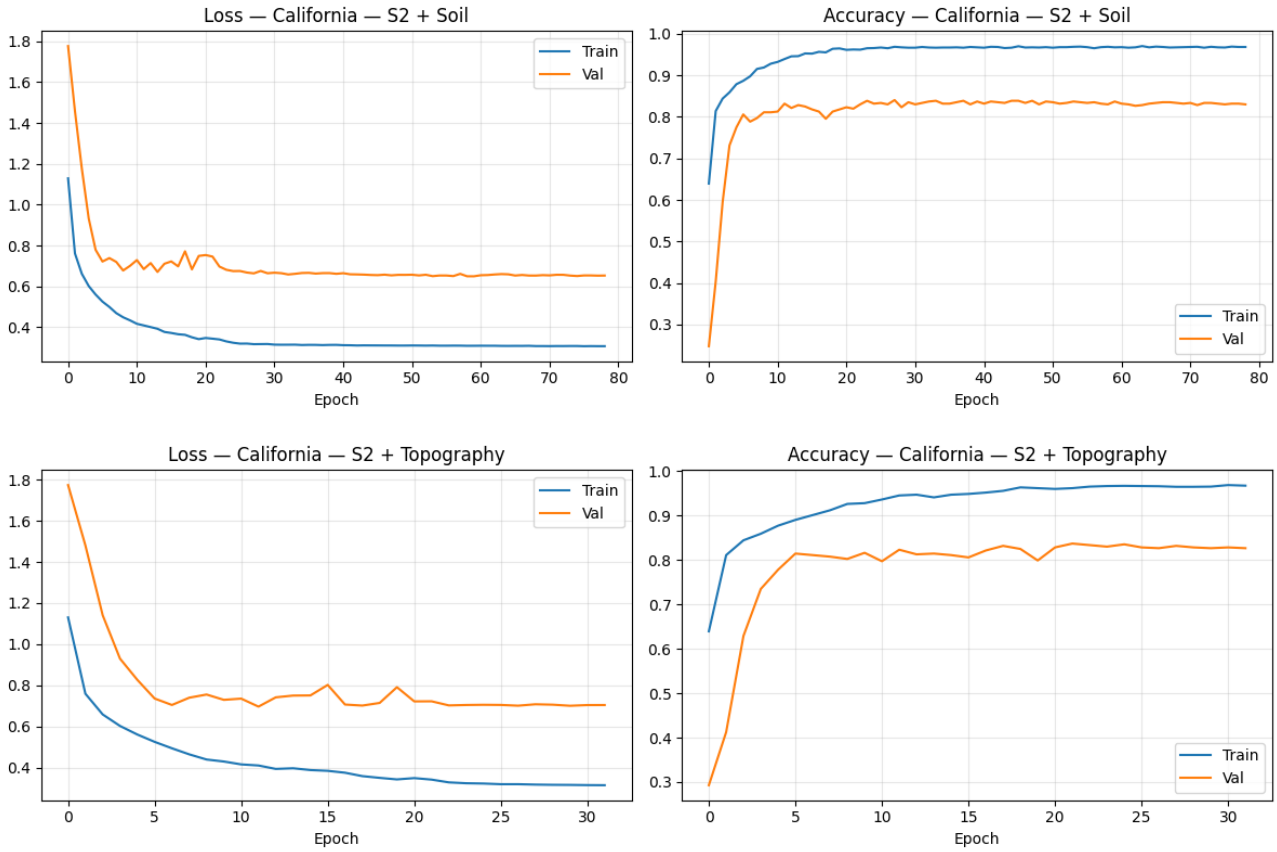


Figure 26: Part 3 convergence curves for California: S2 + Soil (top) and S2 + Topo (bottom).

California: Part 3 Convergence Curves (+All)

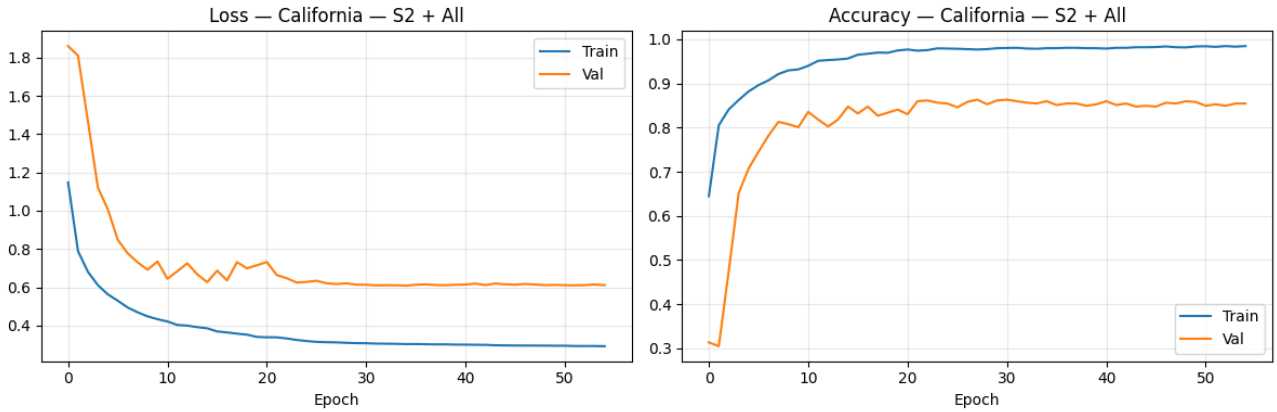


Figure 27: Part 3 convergence curves for California: S2 + All covariates.

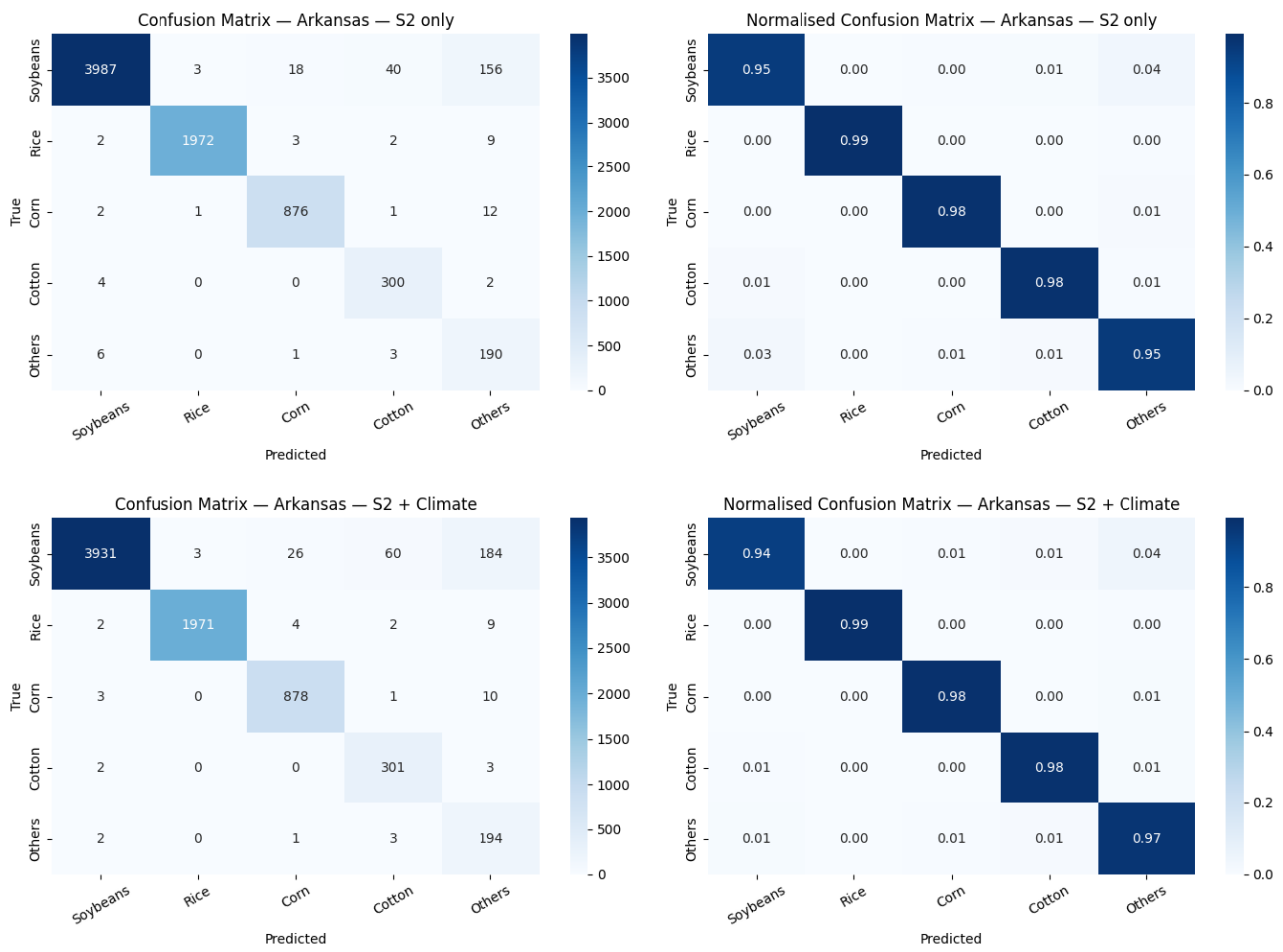


Figure 28: Part 3 confusion matrices for Arkansas: S2 Only (top) and S2 + Climate (bottom).

Arkansas: Part 3 Confusion Matrices (+Soil & +Topo)

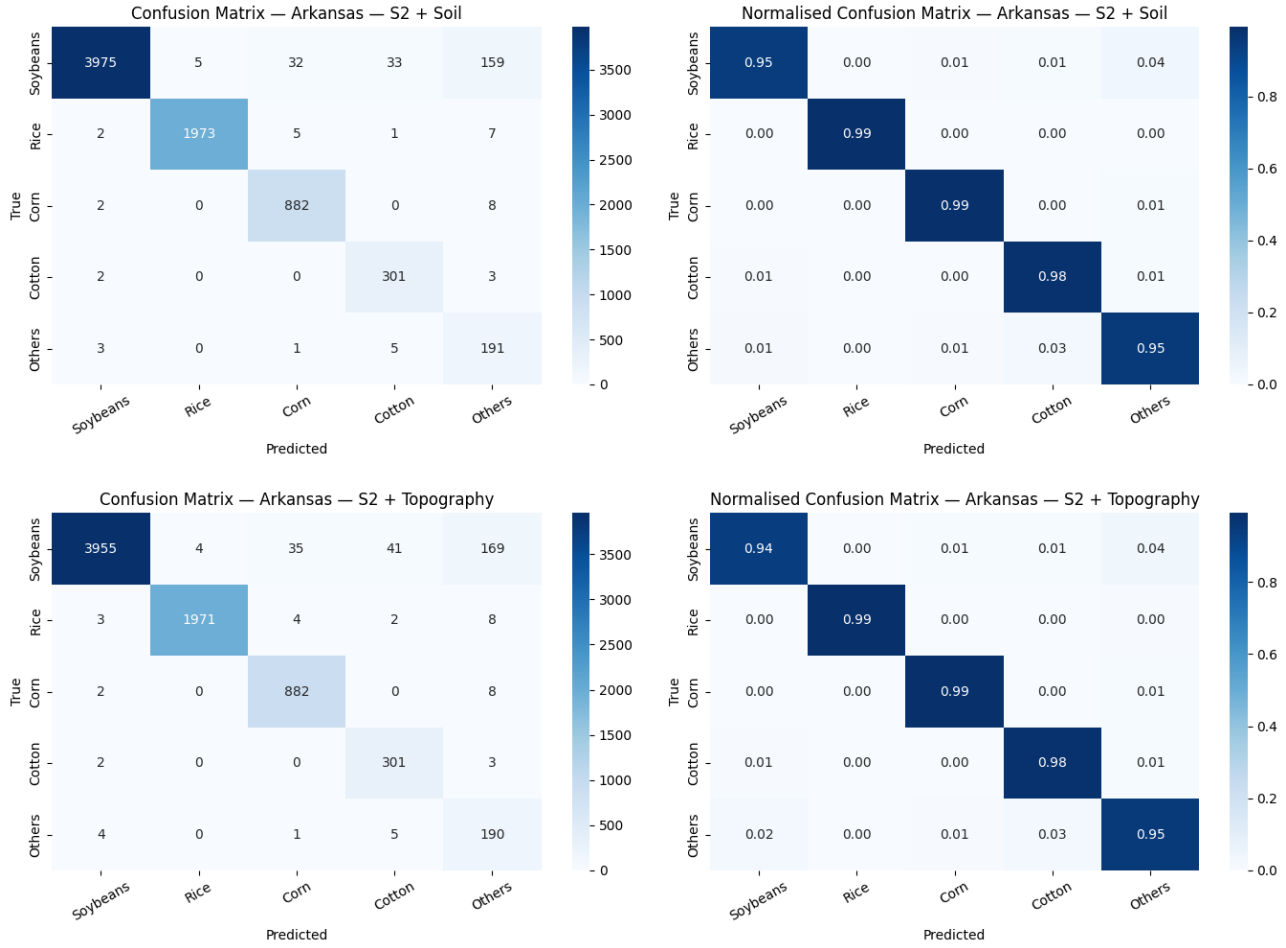


Figure 29: Part 3 confusion matrices for Arkansas: S2 + Soil (top) and S2 + Topo (bottom).

Arkansas: Part 3 Confusion Matrices (+All)

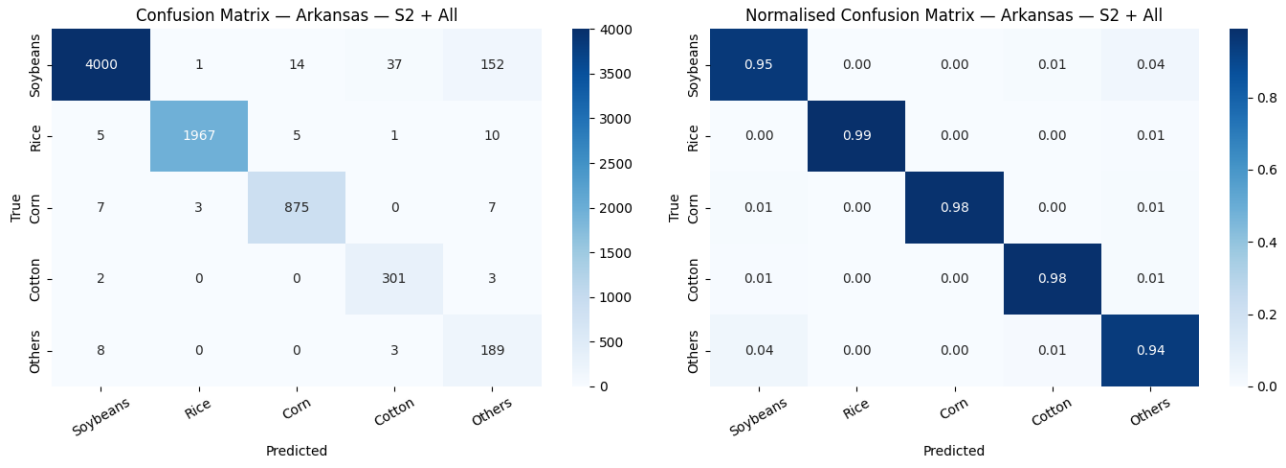


Figure 30: Part 3 confusion matrix for Arkansas: S2 + All covariates.

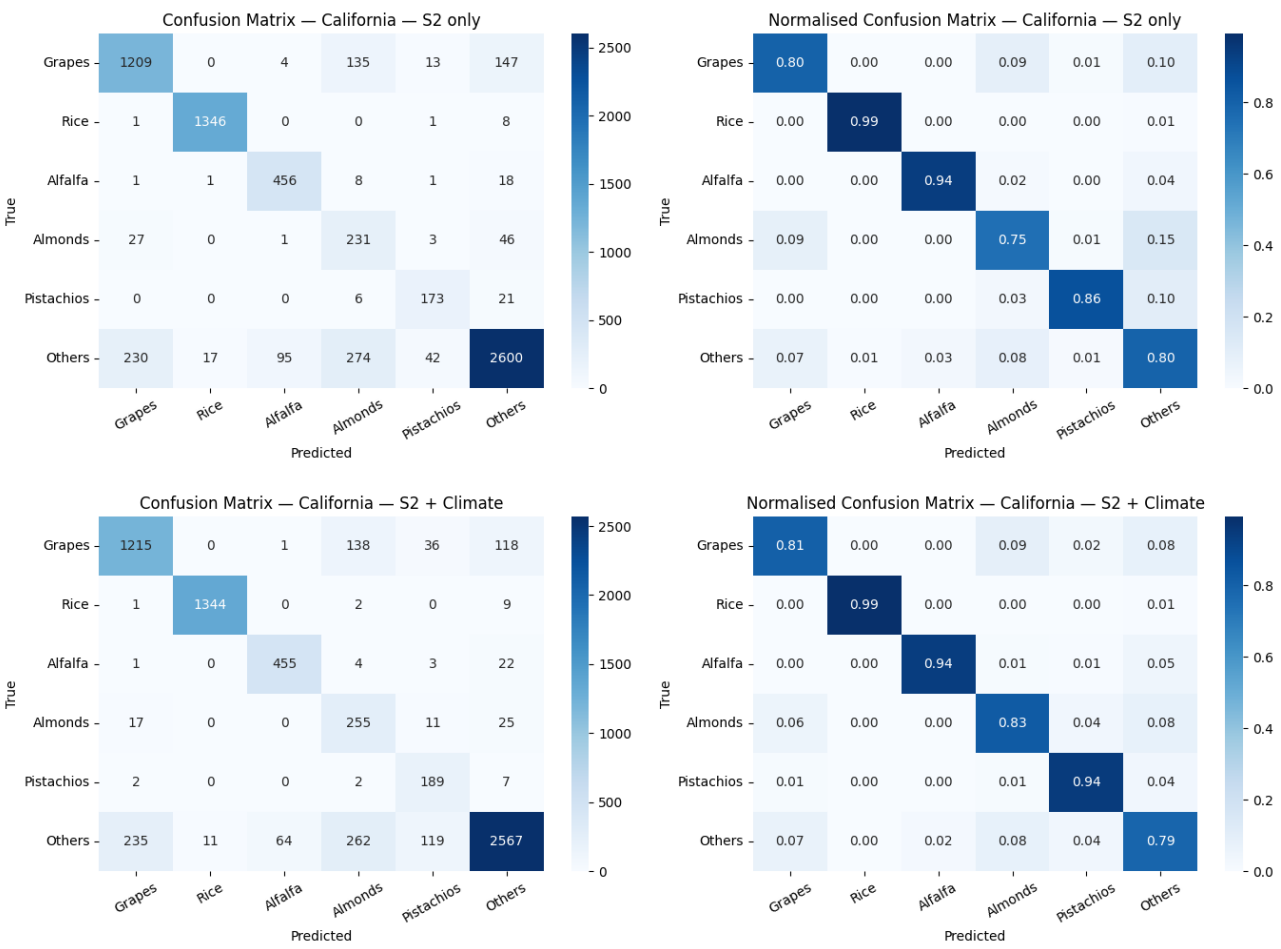


Figure 31: Part 3 confusion matrices for California: S2 Only (top) and S2 + Climate (bottom).

California: Part 3 Confusion Matrices (+Soil & +Topo)

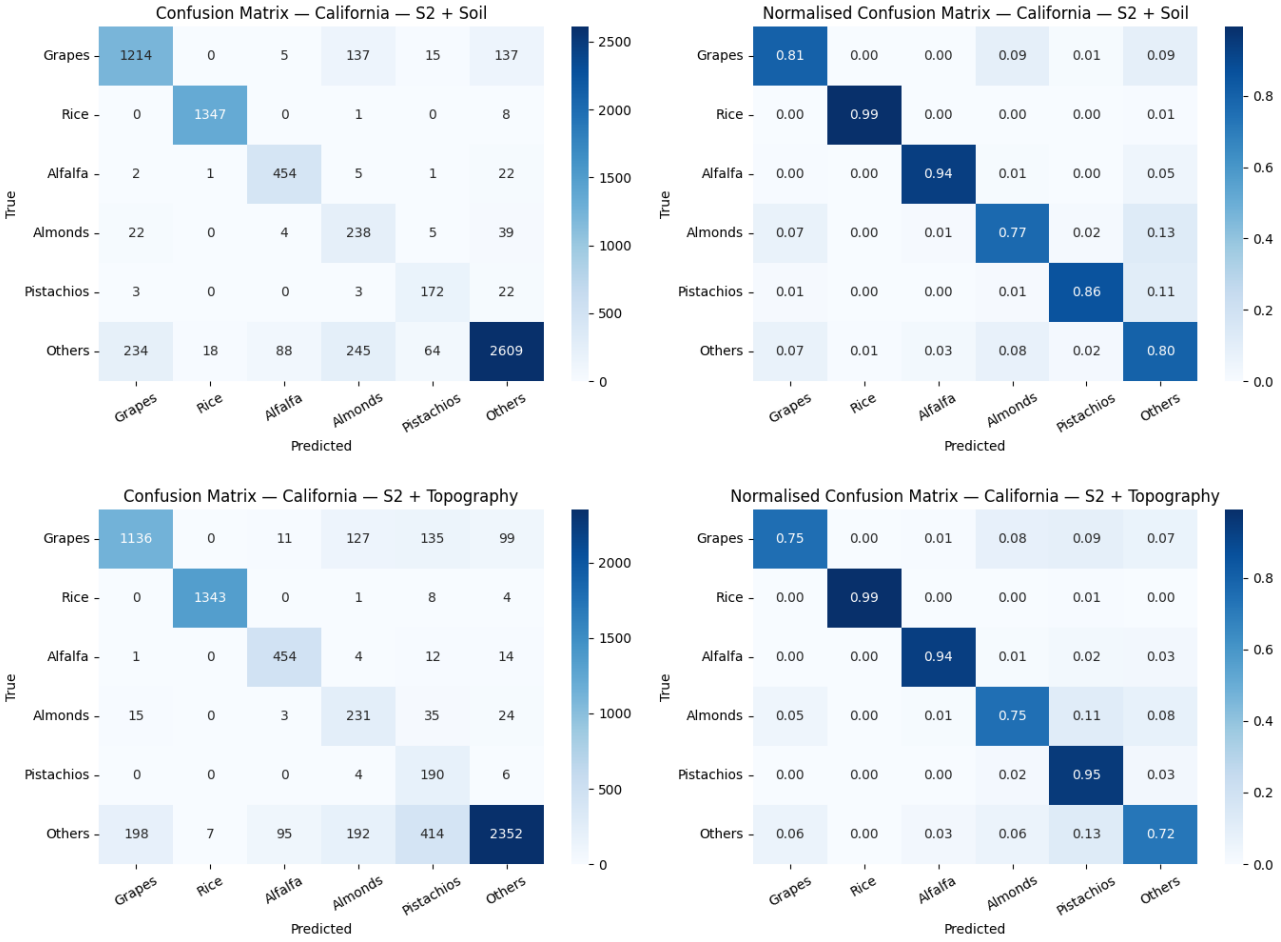


Figure 32: Part 3 confusion matrices for California: S2 + Soil (top) and S2 + Topo (bottom).

California: Part 3 Confusion Matrices (+All)

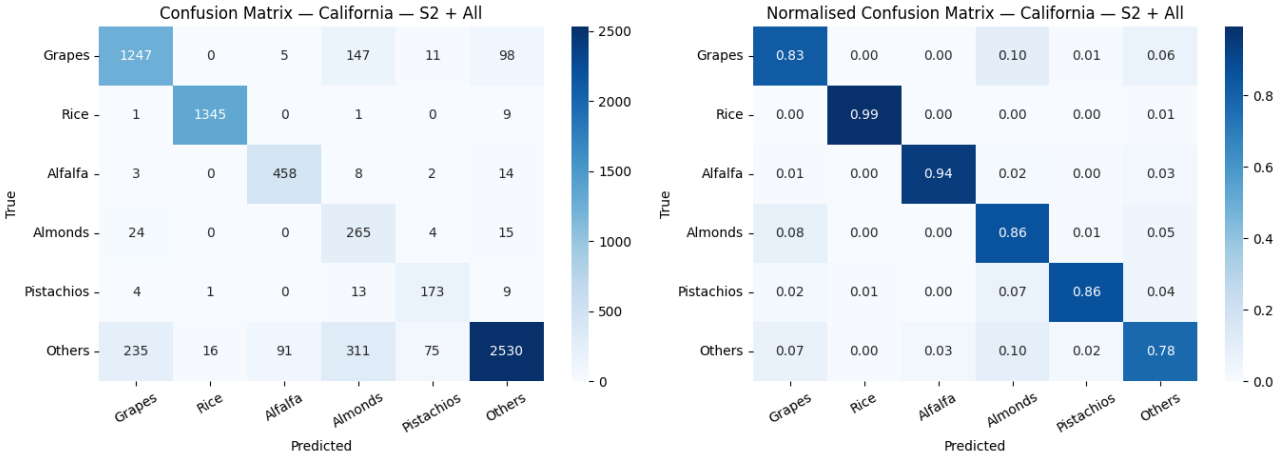


Figure 33: Part 3 confusion matrix for California: S2 + All covariates.

E Covariate Exploration

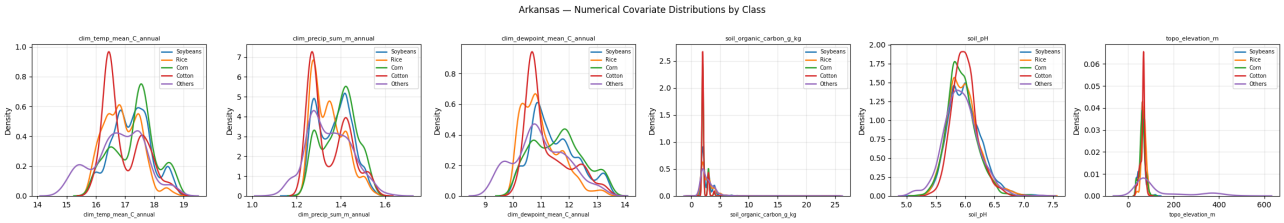


Figure 34: Numerical covariate distributions by class: Arkansas.

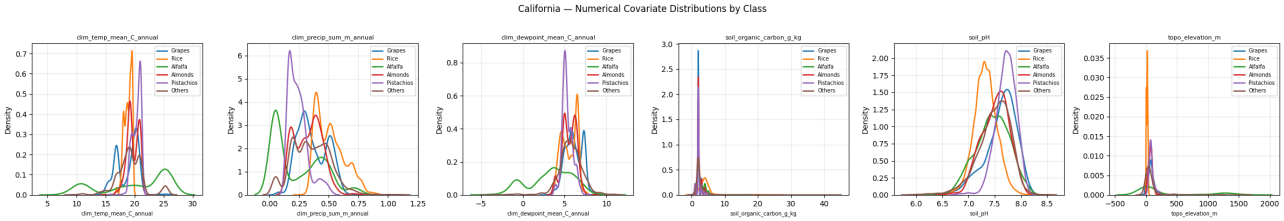


Figure 35: Numerical covariate distributions by class: California.

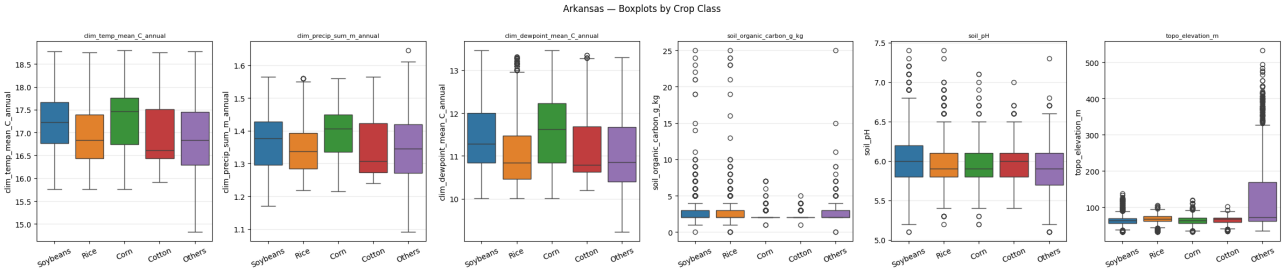


Figure 36: Covariate boxplot distributions by crop class: Arkansas.

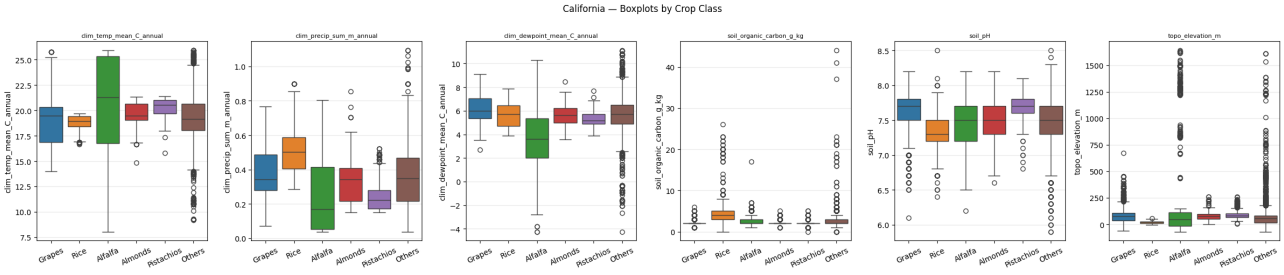


Figure 37: Covariate boxplot distributions by crop class: California.

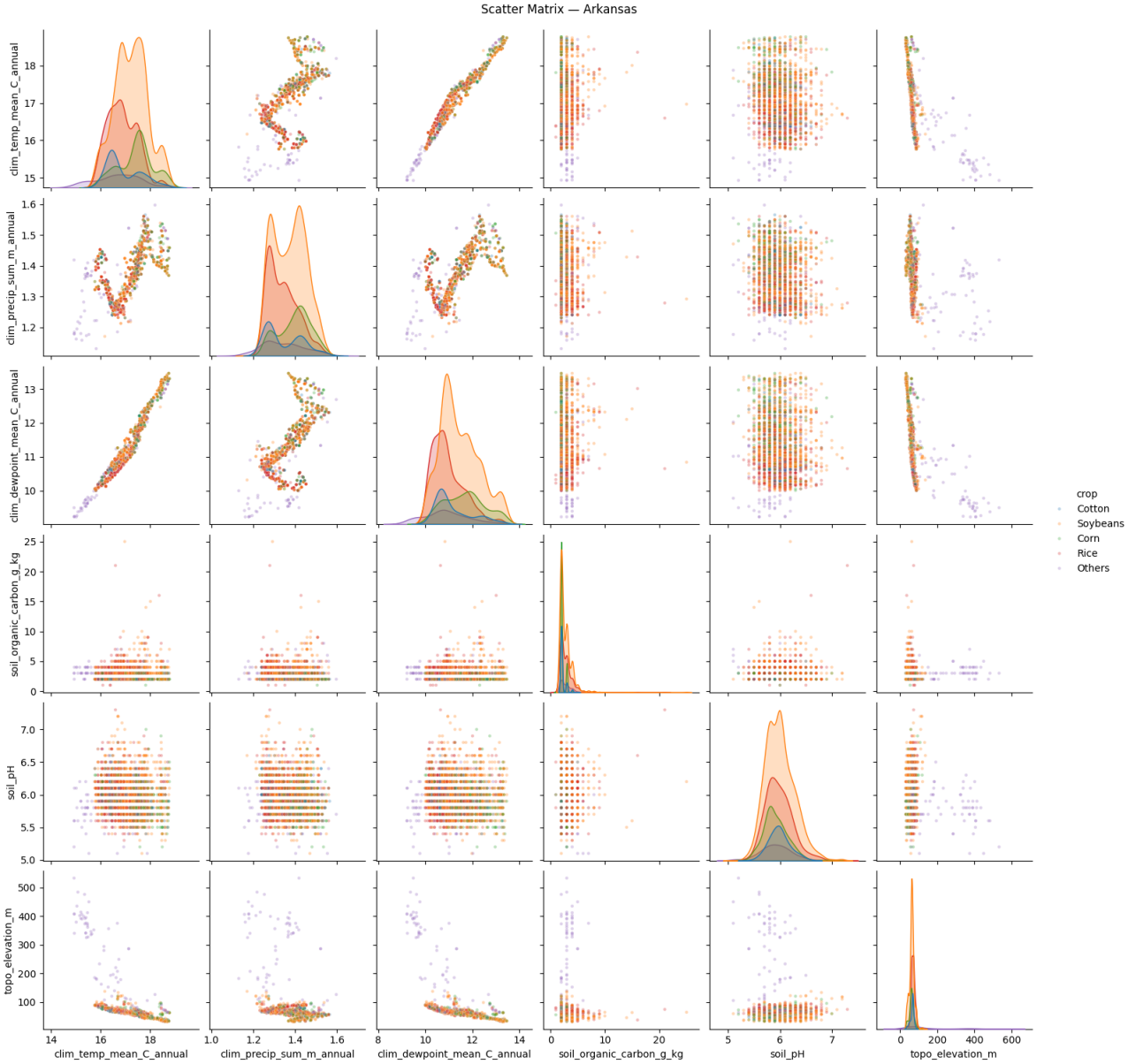


Figure 38: Covariate scatter matrix: Arkansas.

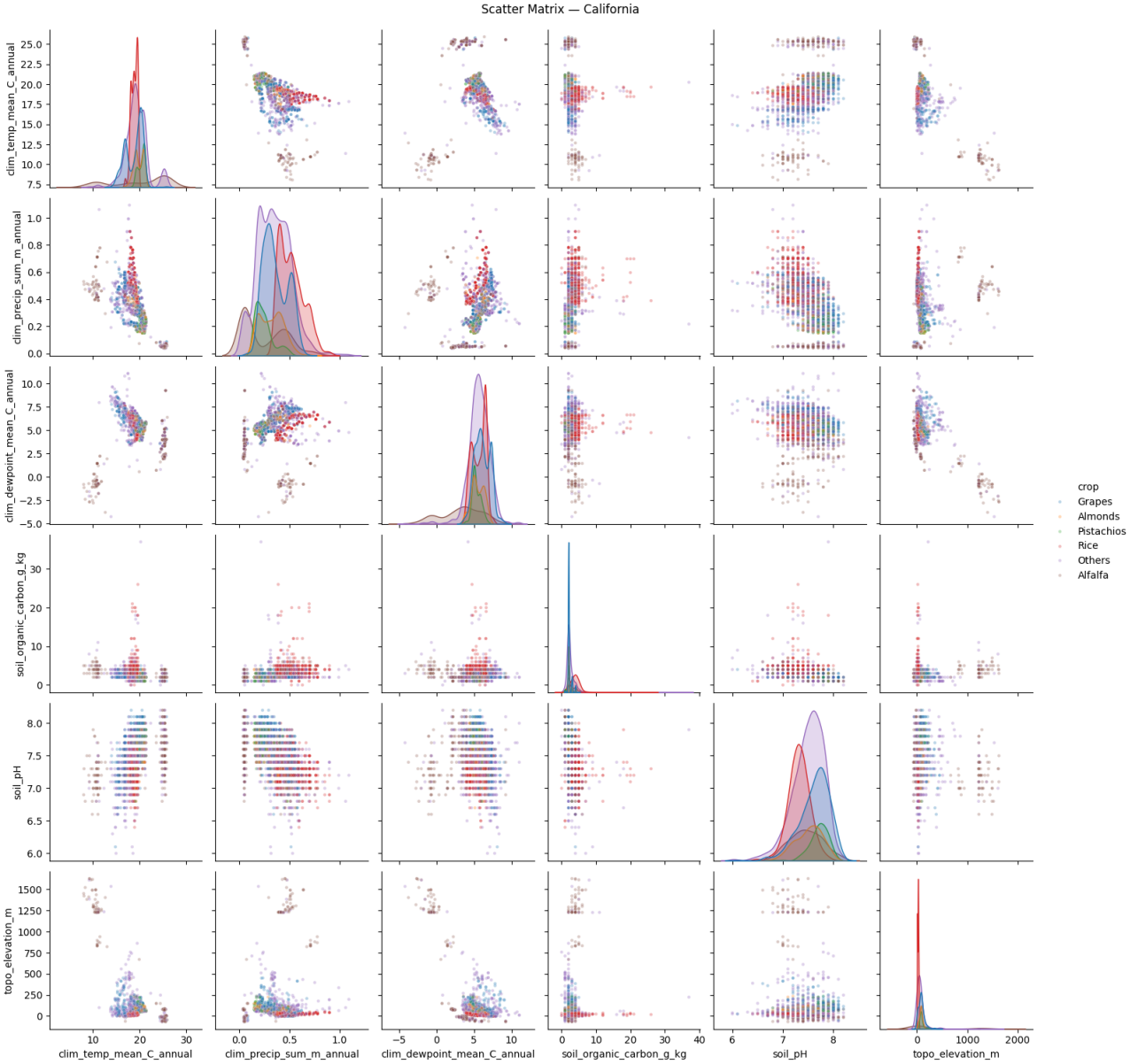


Figure 39: Covariate scatter matrix: California.

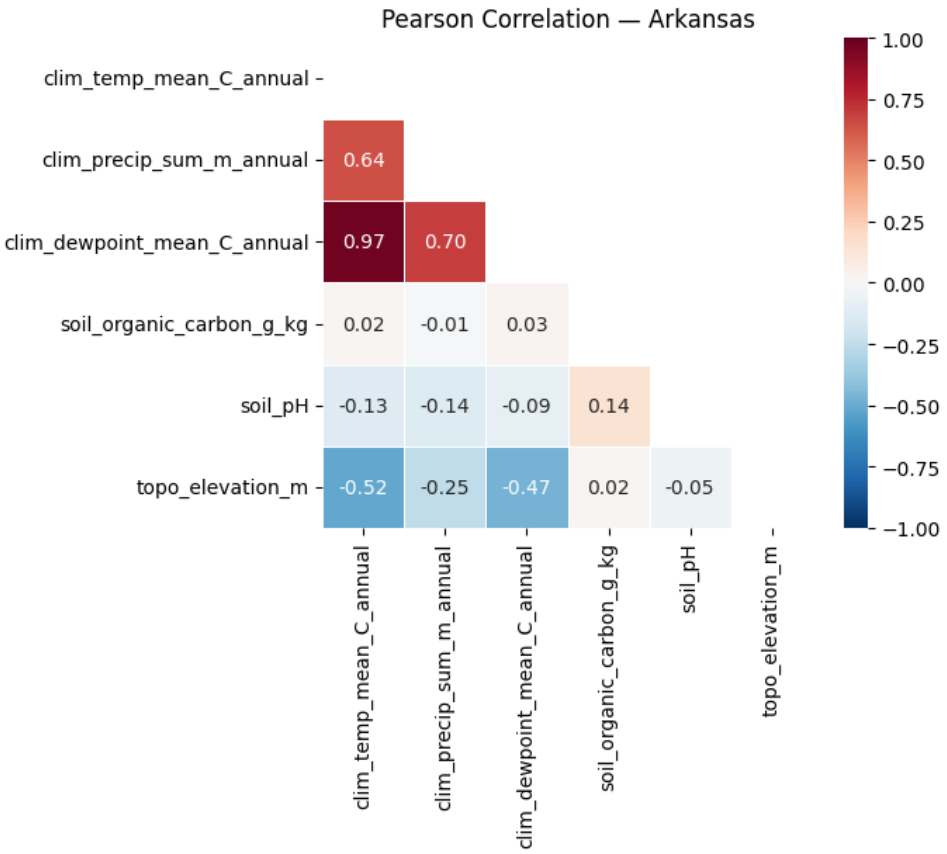


Figure 40: Pearson correlation heatmap: Arkansas.

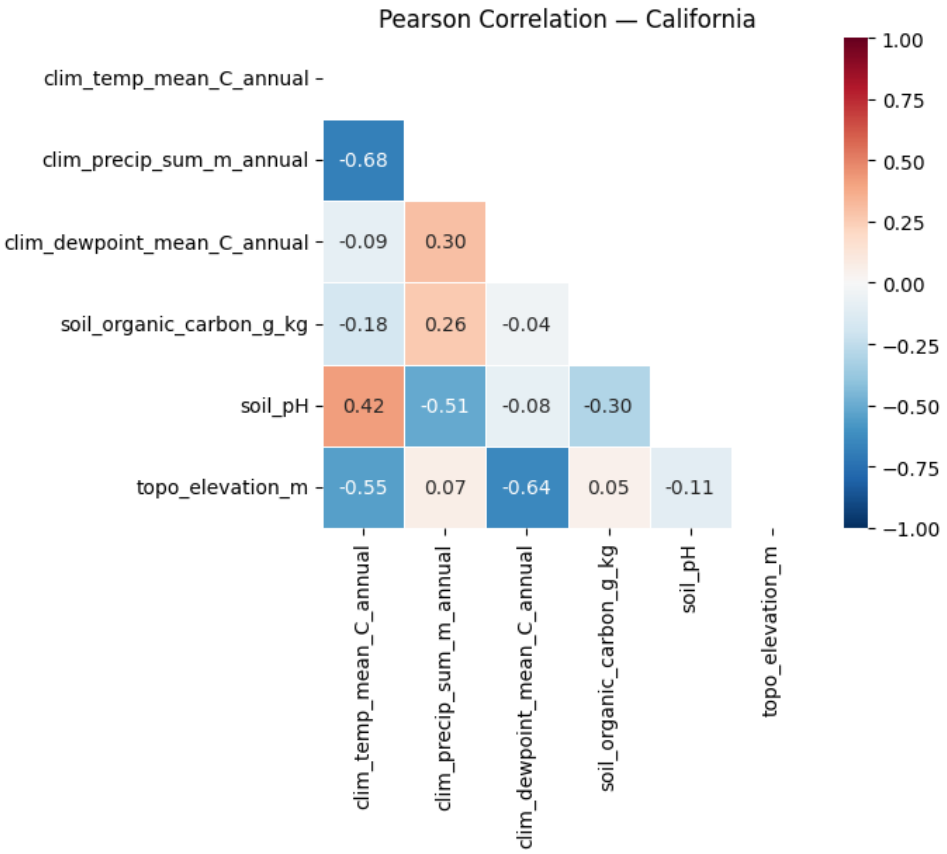


Figure 41: Pearson correlation heatmap: California.

```
===== Arkansas =====

Numerical covariates (ANOVA F-statistic, p-value):
  clim_temp_mean_C_annual                  F=   180.90  p=6.69e-150  ***
  clim_precip_sum_m_annual                 F=   111.68  p=2.72e-93  ***
  clim_dewpoint_mean_C_annual              F=   209.23  p=1.15e-172  ***
  soil_organic_carbon_g_kg                 F=    80.19  p=4.35e-67  ***
  soil_pH                                  F=    19.53  p=5.08e-16  ***
  topo_elevation_m                         F=   721.90  p=0.00e+00  ***

Categorical covariates (Cramér's V):
  soil_texture_class                       V=0.1841  chi2=1356.09  p=3.55e-268
  topo_landform_class                      V=0.0959  chi2=367.77  p=6.55e-61

===== California =====

Numerical covariates (ANOVA F-statistic, p-value):
  clim_temp_mean_C_annual                  F=    75.65  p=4.53e-78  ***
  clim_precip_sum_m_annual                 F=   637.37  p=0.00e+00  ***
  clim_dewpoint_mean_C_annual              F=   574.45  p=0.00e+00  ***
  soil_organic_carbon_g_kg                 F=   379.13  p=0.00e+00  ***
  soil_pH                                  F=   281.61  p=1.03e-282  ***
  topo_elevation_m                         F=   295.93  p=2.87e-296  ***

Categorical covariates (Cramér's V):
  soil_texture_class                       V=0.2692  chi2=3622.76  p=0.00e+00
  topo_landform_class                      V=0.0870  chi2=378.47  p=1.10e-56
```

Figure 42: Covariate-class association analysis (ANOVA F-statistics and Cramér's V) for Arkansas and California.

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Table 11: Project contribution summary. All project components were developed collaboratively by both members.

Project Component	**Contributors**
Part 1	DRINALI Samy Nazim & GHENA Hani
Part 2	DRINALI Samy Nazim & GHENA Hani
Part 3	DRINALI Samy Nazim & GHENA Hani
Report Writing	DRINALI Samy Nazim & GHENA Hani